
PRODUCTBEACON

The State of Government and Law Enforcement Intelligence Markets 2026

Yohay Etsion

Managing Director, ProductBeacon

Build date: 2026-09-14

Disclosure: *This is ProductBeacon's independent, impartial market research on the government and national-security intelligence-software market, based solely on publicly available material. No vendor sponsors, no paywalled data, no analyst-firm reuse. The author, Yohay Etsion, served as VP Product at Cognyte, which competes in fronts covered here; this report reflects the author's personal analysis, not Cognyte's position, and is not investment, legal, tax, or accounting advice. Full independence and sourcing methodology: [Methodology →](#).*

Front 3 – Tactical Intelligence

An independent ProductBeacon market briefing on the government & national-security intelligence software market – field collection, tactical SIGINT, and device forensics. Front 3 of six.

Disclosures. The author, Yohay Etsion, served as VP Product at Cognyte; Cognyte is assessed across this report on publicly available material only, and is **graded in this front (Attention tier)** on its own dated public border-protection page and a dated public tactical-SIGINT deal, with no internal knowledge used. This chapter is independent ProductBeacon research based **solely on publicly available material**; it reflects the author's personal analysis, not any vendor's position. **This report does not constitute investment, legal, tax, or accounting advice.** No claim here should be relied upon as the basis for any investment decision. As of September 2026, the author holds no position in any named public security – directly or via a cybersecurity-sector ETF – and is not compensated by any named vendor.

Front-matter glossary.

- **IMSI / IMEI catcher (cell-site simulator)** – a field device that, per the Electronic Frontier Foundation, “masquerade[s] as legitimate cell-phone towers, tricking phones within a certain radius into connecting to the device.” Its purpose is to collect cellular identifiers (IMSI = the subscriber identity, IMEI = the device/equipment identity), detect a handset's presence, and – on a downgraded link – intercept. *Active* variants transmit; *passive* variants only listen.¹
- **Direction finding (DF) / geolocation** – locating a transmitting device to a physical point using techniques such as Time Difference of Arrival (TDOA – “comparing the signal's arrival time at three or more nodes”) and Angle of Arrival (AoA – “using two direction finding arrays”).⁹
- **Tactical vs strategic interception** – *tactical* collection is field-deployable, proximity-bound, portable, and run against a present target in operational time (this front). *Strategic* collection is standing, mass-scale, network-side collection at national-backbone scale (the National Agencies front). The dividing line is proximity / portability / tempo versus volume / standing infrastructure.
- **Device forensics / extraction** – recovery of digital evidence from a seized device “under forensically sound conditions.” It runs via *logical* (the tool communicates with the operating system through an API), *full-file-system* (privileged “root” access to copy the entire file system and its decryption keys), or *physical* (“a bit-by-bit copy of the physical memory (NAND flash)”) methods. This is the device-in-hand modality of this front.⁵
- **RF-proximity collection** – collection performed by hardware operating on the radio link in the physical vicinity of the target (catchers, DF, Wi-Fi / Bluetooth collection). It is distinct from remote software-implant exploitation, which compromises the device over the network from inside its trust boundary and is carved out of this front entirely (see the Plays section).

3.1 The Playing Ground

Tactical intelligence is the front where collection comes off the network backbone and into the field. Every front above this one collects at a distance — a national agency tapping a carrier core, a law-enforcement unit serving a warrant on a Monitoring Center. This front is different in a single, defining way: the collector is *physically present*. The buyer is a deployed team — a surveillance unit, a counter-terror element, a border or prison authority, a military tactical cell. That team needs to identify, locate, or collect from a target device in the operator's own vicinity, in operational time. The vendors name the property themselves. Rayzone Group markets its tactical line as "last-mile SIGINT capabilities, empowering tactical teams with real-time situational awareness and on-site threat mitigation," delivered through "Tactical RF interception for 2G, 3G, 4G, and 5G networks" and "Designed for stationary, portable, and airborne deployment."² "Last-mile," "on-site," "portable," "airborne" — the vendor's own lexicon names the line that fixes the chapter: *physical proximity / portability plus operational tempo*. That property, and nothing about the underlying radio technology, is what separates this front from the two collection fronts above it.

The category's core sensor is anchored just as cleanly from the other side. The Electronic Frontier Foundation defines a cell-site simulator as a device that "masquerade[s] as legitimate cell-phone towers, tricking phones within a certain radius into connecting to the device rather than a tower."¹ The phrase *within a certain radius* is the boundary itself: this front's sensors are range-limited and locally present, where the National Agencies' collection is standing and remote. The boundary that keeps this chapter coherent has to be stated with unusual precision. The technology rhymes across fronts even as the operational shape does not:

Tactical Intelligence is field-deployable RF / proximity collection against a present target — defined by physical proximity, portability, and operational tempo, not by the radio technology it touches. Remote mobile-implant exploitation (the NSO / Candiru / Intellexa / Pegasus / Predator class) is a technically distinct modality that is carved out of this front and is never graded.

That sentence does the central grading work of the chapter, so it is worth unpacking what it includes and excludes.

What this front IS. Field-deployable RF / proximity collection against a *present* target. The recurring capabilities across the public vendor pages are: cellular-identity collection — IMSI / IMEI and related identifiers — via active or passive catchers; tactical cellular interception; Wi-Fi and Bluetooth identifier collection and exploitation; direction finding and geolocation of a handset to a physical point; and border or checkpoint cellular identification. The form factor is the tell — fixed, vehicular, body-worn, backpack, or airborne — because the system has to go where the target is. Septier, for one, markets a Guardian "Airborne" variant alongside its ground units.³ Sitting adjacent to the RF modalities is a second, increasingly inseparable one: **device forensics at the edge**. Where the target *device itself is in hand* — seized at a border, an arrest, a checkpoint — the tactical act is extraction, not interception, and Cellebrite, MSAB, and Magnet occupy that edge of the front.

What this front is NOT. Three boundaries matter, and the third is the one this chapter is built around. First, it is *not* network-side mass collection. A deep-packet-inspection appliance or a national-backbone passive-collection system sits standing at a carrier core and collects at scale and remotely; this front collects locally, against a present target. The technologies can touch the same cellular protocols, but the defining property is the line: proximity / portability / tempo versus volume / standing infrastructure. A

mass-profiling SIGINT platform is strategic National-Agencies collection even if it ingests data a tactical catcher fed it. Second, it is *not* carrier-mediated lawful interception. A lawful-interception Monitoring Center receives intercept product *through the carrier*, under a warrant, mediated by the LI standards; a tactical system asks no carrier — it stands up its own transient cell or RF sensor in the field and collects directly. The legal trigger and the carrier's involvement are the lawful-interception markers; their absence, plus proximity, is this front's marker.

Third — and this is the line that costs the most to state and the most to get wrong — it is *not* remote mobile-implant exploitation. A mobile-implant product is a **remote software implant**: it compromises a target handset over the network, often with no proximity at all and frequently zero-click, and exfiltrates content from *inside* the device's trust boundary. A tactical system is **RF-proximity hardware**: it manipulates or listens to the radio link near the target and collects identifiers, metadata, location, or — on a downgraded link — content, *from outside* the device. These are distinct modalities. They operate at different points in the kill chain and sell to overlapping but not identical buyers. And — decisively — they are governed by *different* export-control language: intrusion software under Wassenaar Category 4 (Computers) versus IP-network and RF surveillance under Category 5 Part 1 (Telecommunications), a split detailed in the barriers below. I read the carve-out as doing double duty, and I want to be plain that the two halves align by construction rather than by convenience. It is the technically honest boundary — an implant is not a catcher — and it happens to remove from grading exactly the cohort that is most litigation-sensitive. Because the boundary tracks three independent export-control regimes rather than an analyst's preference, drawing it costs the analysis nothing and protects the report. The implant cohort therefore appears only as descriptive market-force commentary in the Plays section, never as a graded contender.

The Tactical Intelligence Boundary & the Carve-Out

GOVERNMENT AND LAW ENFORCEMENT INTELLIGENCE MARKETS 2026 · FRONT 3: TACTICAL INTELLIGENCE

What this front grades — and the export-control split that makes the carve-out technically correct, not convenient

GRADED IN TACTICAL — physical proximity · portability · operational tempo

FIELD RF / PROXIMITY COLLECTION

- IMSI / IMEI catchers (active & passive)
- cellular interception (2G-5G)
- Wi-Fi / Bluetooth identifier collection
- direction finding & geolocation

Rayzone · Septier · Cellxion (wildcard)

listens to / manipulates the radio link near the target, from OUTSIDE the device

DEVICE-IN-HAND EDGE FORENSICS

- extraction from a seized device
- logical / full-file-system / physical
- brute-force unlock sidesteps wire crypto
- kiosk / tablet / ruggedized / drone

Cellebrite · MSAB · Magnet

the target device itself is in hand — seized at a border, arrest, or checkpoint

NEVER GRADED — carved-out modality

REMOTE MOBILE-IMPLANT EXPLOITATION

- a remote software implant, delivered over the network — often zero-click
- exfiltrates from INSIDE the device's trust boundary, no proximity needed

NSO · Candiru · Intellexa · Pegasus · Predator

A technically distinct modality: remote software implant vs RF-proximity hardware.

Descriptive only — no grade, no claim.

THE EXPORT-CONTROL SPLIT THAT ENCODES THE BOUNDARY

WASSENAAR CATEGORY 5 PART 1 (Telecommunications)

IP-network & RF surveillance — the Tactical Intelligence hardware class.
US EAR analog: ECCN 5A001.j (IP-network surveillance systems).

→ catchers, cellular interception, RF monitoring = the graded modality

WASSENAAR CATEGORY 4 (Computers)

Intrusion software — the implant class.
US EAR analog: ECCN 4D004 (delivery of intrusion software).

→ remote mobile implants = the carved-out modality, never graded

• EU Dual-Use Regulation 2021/821, Article 5 — a human-rights end-use catch-all reaching RF/cellular surveillance items even where no list entry applies (EC guidelines 15 Oct 2024).

WHY THE SPLIT MATTERS

The boundary tracks three independent export-control regimes — not an analyst's preference. An implant is a Computers-category intrusion-software control; a catcher is a Telecommunications-category control. That is the public, technically-correct, defamation-safe reason the implant cohort is referenced, not graded.

Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

The graded modalities of this front (field RF/proximity collection + device-in-hand forensics) versus the carved-out remote locked never-graded set and the neighbouring fronts, anchored to the Wassenaar Category 4 / Category 5 Part 1 split. Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

Three buyer misconceptions

“An IMSI catcher lets you listen to the calls.” On a modern network, mostly not. The catcher's native, reliable yield is *identity and presence* — pulling identifiers, detecting and locating a device. Content interception over 4G or 5G requires defeating encryption, which is why the public record describes catchers resorting to a 2G downgrade. Per the EFF, “if the cell-site simulator is able to downgrade the cellular connection to a 2G/GSM connection then it can potentially perform much more intrusive acts such as intercepting call metadata... the content of unencrypted phone calls and text messages.”¹ I read the downgrade dependency as the single most important *technical* fact in this front. As carriers sunset 2G and 3G, the downgrade target narrows, and the classic content-interception value proposition erodes structurally — a force the differentiation axes and the Terrain section both track. The buyer who thinks a catcher is a wiretap is buying yesterday's network.

“Active and passive catchers are the same product with a label.” They are different risk-and-yield profiles. Passive devices, per the EFF, “do not transmit any signals — they work by plucking cellular transmissions out of the air,” while active simulators “broadcast signals that are either stronger than the legitimate cell sites around them, or are made to appear stronger” to force connections.¹ Active catchers emit, which makes them detectable and increasingly *detected*: in March 2025 the EFF released “Rayhunter,” described as “a new open source tool... to detect cellular spying.”⁴ The detectability cuts

directly into product positioning. Several tactical vendors now sell *both* the catcher and its counter, with Rayzone marketing “ArrowCell,” which “detects the presence of an IMSI catcher” and lets users “disrupt and locate their attacker.”² I read passivity as moving from an engineering choice toward a feature buyers pay for.

“Forensics is a lab activity, separate from tactical collection.” In 2026 the public record shows that line dissolving. MSAB states that “the trend in 2026 is moving extraction to the field,” via kiosk, tablet, and express devices,⁵ and Cellebrite markets a “UFED Touch3 Ruggedized Tablet” that “enables comprehensive data extraction collection capabilities anywhere, whether in the lab, a remote location, or in the field.”⁶ I read field-forensics as the *device-in-hand* tactical modality converging with the *device-on-the-air* RF modality. The same deployed unit increasingly wants both, which is why this chapter grades edge-forensics as a part of this front rather than treating it as a separate market.

The data and technology stack

The stack is best read as four modalities the same buyer assembles, unified by the fact that the sensor is physically near the target and portable.

The first and workhorse modality is **cellular-identity collection and tactical interception** — the IMSI-catcher core. An active catcher impersonates a base station to pull devices into connecting; a passive unit listens to the radio environment without transmitting. The public yield is *identifiers*. Septier’s Guardian is publicly described as “capturing up to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi,” extracting “cellular identities, location, and other information from GSM, UMTS, LTE, and 5G devices in its coverage area.”⁷ Rayzone’s “Piranha” is described as “an SDR (Software Defined Radio) collector of mobile phone identifiers (IMSI & IMEI) in 2G, 3G, 4G and 5G environments.”⁸ I read the migration to **software-defined radio** as the quiet architectural shift inside this modality. SDR is what lets a single box claim “2G through 5G” and absorb new-protocol support in software rather than silicon. So “covers 2G–5G” is a positioning claim about software breadth, not a verifiable statement that live 5G-standalone content interception works.

The second modality is **direction finding and geolocation** — turning a signal into a point on the ground. A catcher tells you a target is *present*; DF tells you *where*. The public technical record describes the two dominant primitives: TDOA, which geolocates a signal “by comparing the signal’s arrival time at three or more nodes,” and AoA, “another geolocation technique using two direction finding arrays.”⁹ Tactical vendors fold DF into the cellular line. Septier markets a “Cellular Locator” as “a tactical location tracking solution that provides accurate location information for cellular devices, using advanced direction finders such as Septier Hunter.”⁷ Here a discipline matters that recurs across the front. DF *accuracy* under real field conditions — multipath, urban canyon, single-node versus networked — is the least externally verifiable axis in the chapter, because no vendor publishes meters-of-error under adversarial conditions. The credible public proxy is therefore *form factor* — man-portable versus vehicular array — which is observable. Accuracy is not.

The third modality is **Wi-Fi / Bluetooth proximity collection**. Modern handsets emit far more than cellular, and the same Septier identifier count explicitly spans “those using Bluetooth and Wi-Fi.”⁷ A distinct historical sub-modality targeted Wi-Fi directly: the Wintego “WINT” system was described by named outlets as “a tactical solution for extracting intelligence over WiFi networks, intercepting emails, social networks chats, and web browsing patterns through Man-in-the-Middle (MitM) interception... small enough to fit into a backpack.”¹⁰ (Wintego is named here only as descriptive context for the Wi-Fi-

MitM modality; no current dated self-positioning page surfaces for it, and it is not graded.¹⁰) I read this modality as the one most directly squeezed by platform-level defenses. MAC-address randomization, on by default in modern iOS and Android, erodes the durable-identifier yield the same way the 2G sunset erodes the cellular-content yield.

The fourth modality is **device forensics at the edge** — the device-in-hand case. When the device is seized, the tactical act is extraction. The public record describes a clear technical ladder. *Logical* extraction has “the forensic tool communicates with the OS via API.” *Full file system* has investigators “elevate privileges to gain ‘Root’ access, allowing them to copy the entire file system and its decryption keys.” *Physical* is “a bit-by-bit copy of the physical memory (NAND flash), this is the gold standard of extraction.”⁵ The 2026 inflection is encryption. Per MSAB, “if a device is in a BFU (Before First Unlock) state, the data is heavily encrypted.”⁵ The public answer is compute — “MSAB’s latest innovation allows agencies to utilize high-power GPUs to accelerate the brute-force process; what used to take weeks can now be accomplished in hours.” Cellebrite frames its UFED as “The Industry Standard for Lawfully Accessing and Collecting Digital Data,” telling buyers “don’t let complicated locks, encryption barriers, deleted and unknown content prevent critical evidence from coming to light,” with “full file system (FFS) and physical extractions.”⁶ Magnet’s AXIOM is positioned to “gather all digital evidence associated with a case... whether the evidence lived on a smartphone, the cloud, a computer, a USB drive, other removable media, or even IoT-related devices.”¹¹

I read this sub-front as the *healthiest* part of this front commercially, precisely because it sidesteps the air-interface encryption problem. When the device is in hand and the unlock can be brute-forced, the transport-layer crypto that defeats the over-the-air catcher is bypassed entirely.

The axes buyers discriminate on

Five dimensions separate the serious tactical vendors, each mapping to a real, publicly observable cost-and-capability structure.

1. **RF / protocol coverage, 2G to 5G.** How many cellular generations a system collects against — and the buried distinction beneath the spec sheet, whether “5G support” means 5G *non-standalone* (still anchored to a 4G core) or true 5G *standalone* (where the permanent identifier is encrypted). The public proxy is explicit generation enumeration: Rayzone’s “2G, 3G, 4G, and 5G,” Septier’s “GSM, UMTS, LTE, and 5G,” Piranha’s “2G, 3G, 4G and 5G.”^{2, 7, 8} Every serious vendor now lists 5G. I read uniform “supports 5G” listing as table-stakes positioning, not a capability discriminator. The real, undisclosed differentiator is performance against 5G *standalone* with SUCI encryption, which no vendor quantifies publicly. If, through H2 2026, a tactical vendor publishes a named government reference or an independent test asserting live identity collection against a 5G *standalone* network — not non-standalone — the “5G is just a checkbox” read weakens. Absent that, it holds, and market commentary already flags “5G SA-compatible interception” as the white space “where no vendor has yet established a clearly dominant globally deployed commercial solution.”¹²
2. **IMSI-less / passive identification.** Whether a system can identify or track a target *without actively transmitting*. This matters because active catchers are detectable, now have an open-source detector in Rayhunter,⁴ and trip stricter legal treatment in some regimes for the act of transmitting. The public proxy is explicit “passive” positioning and hybrid active/passive designs. Market commentary notes that “hybrid IMSI catchers, which combine active and passive capabilities, are

--

gaining traction and are expected to be the most lucrative design niche through 2030.”¹⁵ I read the hybrid push as a direct response to detectability and to legal-operational pressure against transmitting — the same force, surfacing as an architecture choice.

3. **DF accuracy and form factor.** How precisely, and from what platform, a system localizes a handset. As the stack section noted, accuracy itself is not outside-verifiable; the credible public proxy is form factor — fixed, vehicular, body-worn, or airborne (Rayzone’s “stationary, portable, and airborne deployment,”² Septier’s airborne Guardian variant³). I read airborne and drone DF as the fastest-moving form-factor story of 2025–2026, because it closes coverage blind spots and reaches denied areas. The *accuracy* claims attached to any form factor, though, stay positioning until tied to a named reference.
4. **Portability and form factor as the defining axis.** This is the axis that *is* the front. A capability delivered in a fixed rack is a strategic or lawful-interception box; the same capability in a backpack, a vehicle, a body-worn kit, or a drone payload is tactical. The public vendor lexicon foregrounds it relentlessly — “last-mile,” “tactical teams,” “portable,” “airborne,” “fits into a backpack.”^{2, 3, 10} I read form factor as the cleanest *public* discriminator in the entire front. Unlike DF accuracy or true 5G-standalone performance, it is visible in the marketing imagery and the spec sheets and cannot be faked.
5. **Encryption compressing classic passive interception — the structural headwind.** This is the defining macro-force on the front, and it works through two compounding effects. First, *transport encryption* on 4G and 5G makes over-the-air *content* interception unreliable without a 2G downgrade,¹ and 2G/3G sunset is steadily removing the downgrade target. Second, *5G SUPI/SUCI encryption* — the standard “attempts to fix IMSI-catching issues by encrypting the permanent identifier (now SUPI) and transmitting the SUCI,” though “risks... remain through downgrade attacks and implementation flaws”¹⁴ — erodes even the *identity* yield over time. The classic passive content-interception value proposition is structurally shrinking as encryption hardens and legacy networks retire. I read this as the force pushing the whole front in three directions at once. It pushes toward active and hybrid catchers that *manufacture* a downgrade; toward device forensics, where brute-forcing the unlock sidesteps transport crypto entirely; and toward DF, metadata, and identity value over content value. If, through H2 2026 and into 2027, vendors’ public roadmaps and named-outlet coverage keep foregrounding forensics, DF, identity, and 5G-standalone research over classic passive-content interception, the compression read holds. If a vendor publicly demonstrates durable 5G-standalone content interception at scale, it weakens.

The public barriers

This front’s defining barrier is not technology — it is export control, which is also the legitimacy frame for the entire surveillance-vendor discussion. Two structural barriers sit on top of that, and naming the regimes *precisely* is itself a discriminator. The intrusion-software line is exactly where the carved-out mobile-implant class lives, and conflating it with RF surveillance would blur the front’s boundary.

Export-control regimes — the defining public barrier. Tactical surveillance hardware is dual-use, and three overlapping regimes gate who may sell it across borders. The Wassenaar Arrangement controls the two relevant modalities under *different* categories, and the split maps exactly onto the carve-out. Per a NIST summary, “the Intrusion Software and IP Network Surveillance controls represent two distinct types of definitions under different sections of the Control List, with the former under Category 4 (Computers)

--

and the latter under Category 5 Part 1 (Telecommunications).¹⁵ Intrusion software — the mobile-implant / Category 4 class carved out of grading — “is defined as software designed to covertly gain access to a computer or other networked device and, once inside, to extract or modify data or modify the execution path.”¹⁵ IP-network and RF surveillance — closer to this front’s graded modalities — sit under Category 5 Part 1. The December 2024 Plenary “adopted updated controls on intrusion software and IP-network surveillance systems.”¹⁵ I read the Category 4 / Category 5 Part 1 split as the cleanest *official* articulation of the front’s boundary itself: an implant is a Computers-category intrusion-software control; a catcher is a Telecommunications-category control.

The EU layer adds a human-rights end-use test on top of the list. Under Dual-Use Regulation 2021/821, Article 5 “mandates exporters to seek authorization if they know or suspect that their cyber-surveillance items could be used for human rights abuses, even if these items aren’t explicitly covered by existing export controls.”¹⁶ Cyber-surveillance items are defined to reach dual-use items “specially designed to enable the covert surveillance of natural persons by monitoring, extracting, collecting or analysing data from information and telecommunication systems.”¹⁷ I read the Article 5 catch-all as the regime most directly aimed at *this* front. It reaches RF and cellular surveillance items even where no specific control entry applies, which is why an EU-headquartered tactical vendor carries a due-diligence burden that is itself a competitive and reputational variable.

The US tracks the same split. EAR ECCN 5A001.j controls “IP network communications surveillance systems or equipment,” capturing “[a]nalysis at the application layer,” extraction of “selected metadata and application content,” and mapping “the relational network of an individual or of a group of people.”¹⁸ ECCN 4D004, by contrast, controls “‘software’ ‘specially designed’ or modified for the generation, command and control, or delivery of ‘intrusion software.’”¹⁸ The 5A001.j-versus-4D004 pairing mirrors the Wassenaar Category 5 Part 1 / Category 4 split — the same boundary, encoded in three independent regimes rather than asserted by this report. Throughout, one discipline binds: any control-classification claim about a named vendor cites the regime text or the vendor’s own statement, never an analyst assertion. A wrong classification on a named company is a credibility *and* a defamation hit.

Domestic-use law as a demand-shaping barrier. Export control governs cross-border *sale*; domestic interception law governs *use*, and it is a real demand gate — a buyer that cannot lawfully deploy will not buy. The US public record is the clearest. Department of Justice policy and proposed legislation treat catcher use as warrant-gated and pen-register-scoped. Cell-site simulators “must be configured as pen registers, and may not be used to collect the contents of any communication, in accordance with 18 U.S.C. § 3127(3),”²⁰ and the bipartisan “Cell-Site Simulator Warrant Act” was reintroduced in November 2023 to “create clear warrant requirements for use of cell site simulators by any government agency.”¹⁹ I read the “configured as a pen register, no content” constraint as the legal codification of the technical reality this chapter opened with — the catcher’s reliable yield is identity and metadata, not content. The law and the physics point the same way, and together they narrow the legitimate use case toward *identify and locate* rather than *intercept content*. Outside the US, each EU member state runs its own interception-authorization regime, and comparable per-state public detail is not uniform. So this chapter asserts the US warrant frame and flags non-US domestic-use law as a thin-material area rather than manufacturing a universal rule.

3.2 The Terrain

Market size: four named markets, walled, never averaged

There is no clean public dollar figure for the tactical field-collection software slice. The honest way to size this front is to set the public estimates of the *adjacent* markets side by side and let the spread between definitions speak. The defining problem here is the sharpest in the report: no research publisher sells a methodology-disclosed “tactical SIGINT / field-collection software” figure. What exists publicly are four adjacent markets, each scoped differently, each bleeding into either carrier-mediated lawful interception, network-side mass collection, or general-purpose forensics. They are walled below and deliberately **not summed** – summing would multiply-count the vendors (Cellebrite, MSAB, the catcher makers) that appear in more than one definition. All figures are research-publisher numbers and are directional; none should be quoted as authoritative.

The first market is **lawful interception**, the broadest published figure that touches this terrain. It is dominated, though, by carrier-mediated, warrant-scoped interception, which is the Monitoring Centers’ domain, not field collection. It is carried only as the *outer envelope*, the largest published “interception” pool, with the explicit caveat that the tactical slice is a small, opaque sub-fraction of it.

Publisher	Definition used	Figure (as stated)	CAGR / period
Research and Markets	Lawful interception solutions/services across network types	USD 6.24B (2024) → USD 7.50B (2025) [June-2026 edition]; re-issued Sept 2026 edition: USD 3.98B (2026) → USD 6.98B (2032) ²¹	~20.12% (2025–2032) [June-2026 ed.]; 9.47% (2026–2032) [Sept-2026 ed.]
MarketsandMarkets	Lawful interception by network, component, type	USD 8.20B (2026) → USD 24.60B (2032) ²²	20.1% (2026–2032)
Mordor Intelligence	Lawful interception market	USD 5.47B (2025) → USD 6.47B (2026) → USD 14.98B (2031) ²³	18.27% (2026–2031)

On their June-2026 editions the three publishers bracketed the 2025 lawful-interception market at roughly USD 5.5B–7.5B with CAGRs in the high-teens to low-twenties. Research and Markets’ September-2026 re-issue drops its base to USD 3.98B (2026) at a 9.47% CAGR, so the current-edition bracket is wider (roughly USD 4.0B–8.2B for 2026) and the CAGR cluster no longer tight. I read that tight CAGR cluster, on the June-2026 editions it was drawn from, as evidence the *interception* category overall is genuinely fast-growing. The dollar level, though, is the Monitoring Centers’, not this front’s, because lawful interception as published is the warrant-scoped business of the Law Enforcement Monitoring Centers front. The binding caveat: this is the outer interception envelope, and field / proximity collection is an un-isolated sliver of it. It is not this front’s market.

The second market is **IMSI-catcher and cellular interception**, the tightest public fit to this front’s field-collection core – RF-proximity collection against a present cellular target. It is also far smaller than lawful interception, consistent with the narrower field-hardware scope.

Publisher	Definition used	Figure (as stated)	CAGR / period
MarketIntelto	IMSI catcher market (devices mimicking cell towers for ID/location/interception)	USD 400M (2024) → USD 950M (2033) ²⁴	9.8%
Mordor Intelligence	Technologies/solutions to “intercept, monitor, and analyze cellular communications” for LEA/intelligence/government	USD 695.49M (2025) → USD 735.69M (2026) → USD 921.57M (2031) ²⁵	4.61% (2026–2031)
Mordor Intelligence (IMSI-catcher variant)	IMSI catcher market	USD 182.57M (2025) → USD 207.68M (2026) → USD 306.47M (2031) ²⁶	8.08%

This is where the definitional incoherence is starkest, and the spread is the finding. The “cellular interception” framing brackets 2025 at roughly USD 695M; the narrower “IMSI catcher” framing brackets 2025 at roughly USD 183M; a third single-firm IMSI-catcher framing (MarketIntelto) puts the 2024 base at roughly USD 400M. That is a **roughly four-fold spread (USD 183M to USD 695M) for ostensibly the same field-collection category**, turning entirely on whether “cellular interception” pulls in software and services or only catcher hardware. I read the spread itself as the result. The field-collection software slice that this front’s contenders actually sell sits somewhere inside the USD 183M–695M envelope, but no publisher isolates it cleanly, and the publishers disagree by nearly four-fold on where the boundary even falls. Notably, the CAGRs here (4.6%–9.8%) run *well below* the lawful-interception category’s roughly 20% – consistent with the encryption-driven compression of classic passive cellular interception that the Playing Ground section named. If a methodology-disclosed publisher ever isolates “tactical cellular-collection software” and lands outside this USD 183M–695M envelope, the bracketing is weaker than I have assumed.

The third market is **mobile and digital forensics**, the device-in-hand modality – and the richest published material in the entire front, because three of the leading vendors are public or public-ish (Cellebrite, MSAB, Magnet under Thoma Bravo). This is the one definition of this front with filing-grade vendor anchors.

Publisher	Definition used	Figure (as stated)	CAGR / period
MarketsandMarkets	Mobile forensics (extraction/analysis of data from mobile devices)	USD 5.72B (2026) → USD 9.99B (2031) ²⁷	11.8% (2026–2031)
Precedence Research	Digital forensics (mobile forensics stated to dominate revenue share in 2025)	USD 13.46B (2025) → USD 15.50B (2026) → USD 53.65B (2035) ²⁸	14.83% (2026–2035)
Mordor Intelligence	Digital forensics market	USD 14.85B (2025) → USD 31.74B (2030) [pre-July edition]; re-issued 2026-07-23: USD 7.51B (2025) → USD 8.46B (2026) → USD 15.37B (2031) ²⁹	16.5% (2025–2030) [prior ed.]; 12.68% (2026–2031) [July-2026 ed.]

The *mobile-forensics* sub-definition brackets 2026 at roughly USD 5.7B; the broader *digital-forensics* parent brackets 2025 at roughly USD 7.5B–13.5B on the current editions, with mobile forensics stated to be the dominant slice, and CAGRs cluster in the low-to-mid-teens. Mordor's 2026-07-23 re-issue states a materially lower 2025 base – USD 7.51B against the USD 14.85B of the edition this report originally cited. I read mobile forensics as the single best public proxy for this front's device-extraction modality specifically – and it is the *only* definition adjacent to this front with a public-issuer anchor to validate it. The scope caveat: digital forensics also includes computer, cloud, and network forensics that are not field-tactical, so the mobile-forensics sub-figure (roughly USD 5.7B, 2026) is the tighter fit. One rare public concentration datapoint sits here as market structure. MSAB's 2025 annual report states that “the top five vendors, OpenText, Cellebrite, Magnet Forensics, Exterro, and MSAB, accounted for about 38% of 2025 revenue” in digital forensics.^{29 30}

The fourth market is **signals intelligence**, the relevant anchor for the military-tactical end of the front. It is dominated, though, by strategic, airborne, and naval SIGINT systems (the National Agencies' and the defense primes' domain), far broader than tactical field-collection software. It is carried as a demand-envelope ceiling only.

Publisher	Definition used	Figure (as stated)	CAGR / period
Fortune Business Insights	Signals intelligence market	USD ~19.84B (2025) → USD 29.77B (2034) ³¹	4.61%
Future Market Insights	Signals intelligence (SIGINT) market	USD ~17.7B (2025) → USD 30.0B (2035) ³²	5.4%

The SIGINT publishers bracket 2025 at roughly USD 17.7B–19.8B with low-to-mid-single-digit CAGRs (4.6%–5.4%) – a *much* slower growth profile than forensics or lawful interception, consistent with a mature, prime-dominated, hardware-platform market. I read SIGINT as the strategic / military *ceiling* on the addressable pool, not a tactical-collection-software size; tactical field SIGINT is an un-isolated sub-fraction. A pair of public budget figures fixes the demand envelope without being mistaken for a software market. The US Army allocated roughly USD 95.4M to tactical communications modernization, and US DHS allocated roughly USD 43.8B for surveillance in FY2026.²⁵ I carry both strictly as demand-envelope proxies, far broader than the tactical-collection slice, never as a market size.

The tactical-collection slice: a labeled proxy, not a segment size

No clean public dollar figure isolates the tactical field-collection software slice. Given the deliberate opacity of this category, where much of the hardware is sold under classification or export control and the vendors are mostly private, that thin-material result is *expected and honest*, not a gap to be filled by invention. The four definitions above bracket adjacent markets that overlap heavily and must not be summed. Set side by side, they read as four walls. Lawful interception (roughly USD 5.5B–7.5B for 2025 on the June-2026 editions; roughly USD 4.0B–8.2B for 2026 on the current editions) is the carrier-mediated outer envelope, mostly the Monitoring Centers. IMSI-catcher / cellular interception (roughly USD 183M–695M, 2025) is the field-collection core proxy, where publishers disagree by nearly four-fold on the boundary. Mobile forensics (roughly USD 5.7B, 2026) and digital forensics (roughly USD 7.5B–13.5B for

2025 on the current editions, against USD 13.5B–14.9B on the editions originally cited) are the device-extraction modality and the front's richest material. SIGINT (roughly USD 17.7B–19.8B, 2025) is the strategic / military ceiling, mostly the National Agencies and the primes.

Cellebrite and MSAB appear in the forensics figures, the catcher makers appear in the cellular-interception figures, and the military platforms appear in SIGINT. So summing would multiply-count the same companies and the same revenue. The hardest public currency for sizing this front is not any market derivative at all. It is the disclosed financial performance of a public issuer in it.

That issuer is Cellebrite. Its disclosed revenue and ARR are the cleanest filing-grade evidence that tactical and forensic demand is real — carried strictly at the disclosed level, the most recent reported quarter, never as a market size and never extrapolated.

For the three months ended 30 June 2026 (Q2 FY2026), Cellebrite reported total revenue of **USD 131.1M, up 16% year-over-year**, with subscription revenue of USD 119.5M (up 16%). It posted **Annual Recurring Revenue of USD 507.8M, up 21% year-over-year**, dollar-based net revenue retention of 117%, and GAAP gross margin of 80.8%. The prior quarter (Q1 FY2026, ended 31 March 2026, reported 2026-05-14) was total revenue USD 128.3M, up 19%, and ARR USD 493.0M, up 21%, at 82.5% GAAP gross margin and 115% net revenue retention.³³

I read a single vendor running roughly USD 508M ARR, growing 21% year-over-year at 80.8% GAAP gross margin with 117% net retention, as the cleanest public evidence that the *device-extraction* end of tactical demand is real, recurring, and accelerating. That holds even though no publisher isolates the tactical-collection segment, and even as the catcher TAMs grow far more slowly. If Cellebrite's ARR growth decelerates below roughly 12% across two consecutive quarters in 2026, the "accelerating device-extraction demand" read weakens. (Cellebrite trades as NASDAQ: CLBT and the trading-position disclaimer in the front-matter applies; the figure should be re-verified against the next 6-K at authoring rather than carried stale.)

The corroborating second pure-play is MSAB (Nasdaq Stockholm: MSAB B). Its full-year 2025 net sales reached **SEK 461.8M, up 14.1%** (currency-adjusted 19.7%), with operating profit of SEK 67.8M, in what the company called "the most successful year in the company's history."³⁰ Magnet Forensics is the third name but is now private. Thoma Bravo agreed to acquire it for about C\$1.8B (roughly USD 1.3B), announced 20 January 2023, combining it with Grayshift.³⁴ No current revenue is public, and the take-private price is the last filing-grade figure.

Demand drivers

Five forcing functions carry the analytical weight the opaque market numbers cannot.

Encryption compressing passive interception — the structural ground. The root driver across the front is that the spread of end-to-end encryption has compressed the value of classic passive over-the-air interception: WhatsApp, Signal, Telegram, and modern device-OS encryption mean that intercepting the *wire* increasingly yields ciphertext, not content. The practitioner framing is explicit: "the widespread use of end-to-end encryption by apps like WhatsApp, Signal, and Telegram makes data extraction increasingly challenging."³⁵ The policy framing is institutional. The European Parliament's 2025 briefing, "Access to data for law enforcement and digital forensics," casts device-level forensic access as the law-enforcement response when communications interception is defeated by encryption.³⁶ The market consequence is bimodal. Demand shifts *toward the endpoint*, where end-to-end encryption is defeated

at the device rather than the wire — exactly the forensics-extraction demand the public-issuer anchors are capturing — and *toward active* RF collection that forces a target's device to attach, over passive monitoring. I read the encryption shift as the single structural force reshaping this front. It simultaneously compresses the classic passive-interception sub-market and expands the device-extraction sub-market, which is why the forensics vendors are the front's growth story and the catcher figures are flatter. If cellular-interception CAGRs re-accelerate above the forensics CAGRs through 2026–2027, that read weakens.

The device as the richest target — border and arrest extraction demand. The clearest *quantified* public signal of device-extraction demand is at the US border. Customs and Border Protection reported searching **47,000 electronic devices in 2024, up from roughly 8,500 in 2015** — a roughly five-and-a-half-fold increase over a decade,³⁷ the kind of operational-volume growth that drives recurring extraction-tool spend. The procurement follows the volume. DHS/ICE and Homeland Security Investigations planned a five-year IDIQ with a USD 100M ceiling for Cellebrite mobile-device extraction. Between 2008 and April 2025, ICE awarded Cellebrite at least 213 contracts worth over USD 48.6M for UFED technology and related services.^{37 38} I read the border and arrest context as the demand surface where this front's modality is least encryption-defeated — physical possession plus compelled or exploited access — and most recurring, because every seized device is a unit of demand. If device-search volumes plateau or fall through 2026 — a policy reversal or a legal constraint — the recurring-extraction read weakens.

Border, migration, and event-security cellular ID. Beyond device-in-hand extraction, the *field cellular-ID* use case — identifying and locating devices at borders, migration chokepoints, and mass events — is a named driver in its own right. Publishers cite UAV-mounted interception payloads that “extend tactical flexibility and range, particularly for border surveillance,” and network densification (198,100 outdoor small cells in the US by mid-2025) concentrating traffic at fewer aggregation points.²⁵ I read border and event security as the context that keeps the *catcher* sub-market alive even as encryption compresses passive content interception. Cellular *identity* (IMSI/IMEI presence and location) does not require breaking content encryption at all. If catcher procurement migrates wholly to content-interception framing rather than ID and location, this read is over-stated.

The 5G and VoLTE transition forcing a refresh cycle. Every generational network transition obsolesces part of the installed catcher and interception base and forces a refresh. Operators migrating to standalone 5G cores “must expose interception interfaces per ETSI TS 33.127 v18.11.0 standards,” and 181 carriers across 73 nations had invested in such upgrades by late 2025. Mordor's IMSI-catcher report flags 5G NR solutions as the fastest-growing technology segment, at roughly 11.44% CAGR through 2031;²⁶ its cellular-interception report puts 5G NR at roughly 5.36% CAGR on the wider cellular-interception base.²⁵ I read this as a structural, standards-driven demand pulse that favors vendors who can keep pace with the protocol stack. If 5G-standalone interception solutions fail to out-grow the legacy cellular-interception base through 2026–2027, the refresh-cycle read weakens.

Accountability and oversight pressure reshaping who can buy and sell. The countervailing force is oversight and export control: EU dual-use controls explicitly capturing IMSI-catchers as cyber-surveillance items, US EAR cybersecurity-item controls, and sustained press and NGO scrutiny of border device searches. I read accountability pressure not as shrinking demand so much as *reshaping the supply side*. It concentrates sales toward vendors who can clear export licensing and survive public scrutiny, and pushes the most litigation-sensitive modality (remote mobile exploitation, the carve-out cohort handled in the Plays section) toward the regulatory and reputational margins. If a major vendor on

this front is materially constrained by a named export-license denial or sanction through 2026, the “oversight reshapes supply” read strengthens — but any such claim is named-outlet-attributed or dropped.

The procurement structure

This front’s procurement is bimodal and uniquely export-control-gated. At the field-collection (catcher) end it tracks opaque sovereign and police buying constrained by dual-use export licensing. At the device-extraction (forensics) end it runs through the same accreditation-and-vehicle federal lane as the rest of government software, but with named, datable contract awards. Both modes are public-anchorable.

Export licensing as the gating barrier. For tactical RF-collection hardware, export control is the single richest public table-stakes barrier — and, as the Playing Ground section detailed, the legitimacy frame for the surveillance discussion. IMSI-catchers and cellular-interception equipment are explicitly captured as cyber-surveillance items under both the Wassenaar Arrangement and the EU framework. EU guidance states that controlled cyber-surveillance items include “equipment for covertly extracting communication content and metadata that is transmitted over the air via wireless communication, and radiofrequency monitoring equipment.”³⁹ Annex I of EU Regulation 2021/821 implements the regime controls into the EU’s binding dual-use list, and Article 5 adds the catch-all for items not on the list where there is awareness of potential misuse.⁴⁰ The US implemented complementary EAR controls on cybersecurity items in March 2022.³⁹ I read export licensing as the *real* barrier to entry at this front’s field-collection end. It is a regime-gated, licence-by-destination gate that constrains *who can sell to whom* independent of product quality, and the public lever through which accountability pressure reshapes the supply side. A named export-license denial or sanction materially constraining a major vendor on this front through 2026 — named-outlet-attributed — would confirm the export-control-as-moat read. (Specific ECCN classifications are not asserted per vendor here; that is a defamation and credibility surface, and the regime, not a vendor-specific classification, is what is cited.)

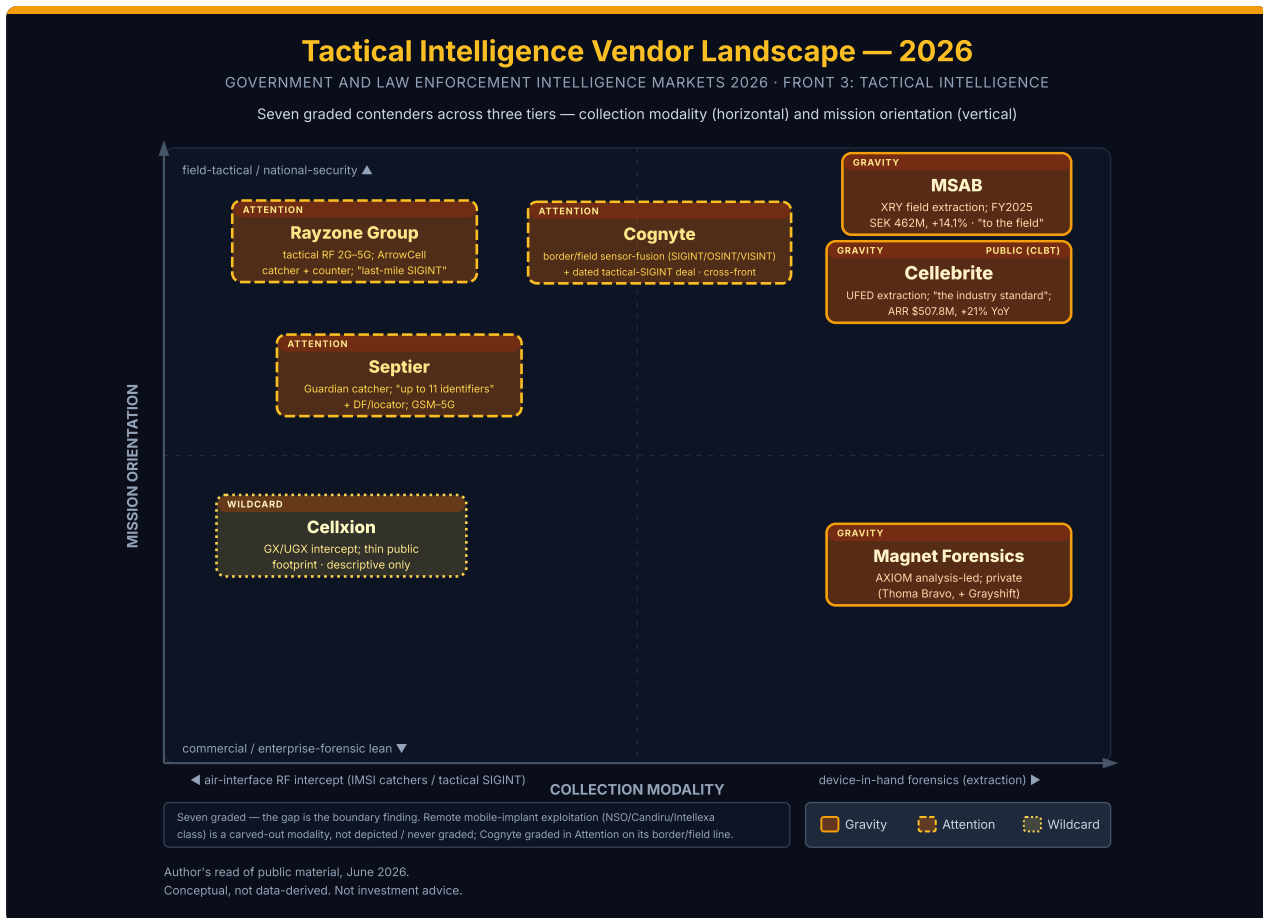
Federal vehicles and named, datable device-extraction awards. At the device-extraction end, procurement runs through established federal vehicles and produces named, public awards — the cleanest procurement evidence in the front. The FBI and DOJ awarded Cellebrite a contract valued at USD 5,717,319 for unlock and extraction software, starting 30 September 2025 and completing 29 September 2030, through the NASA SEWP vehicle.⁴¹ DHS/ICE and HSI planned the five-year USD 100M-ceiling IDIQ for Cellebrite, and CBP had eight Cellebrite contracts set to end between July 2025 and April 2026.³⁷ Vendors market availability through government-wide vehicles as a sales motion in its own right, and MSAB has publicly disclosed a US government-agency order.⁴² I read the device-extraction procurement lane as the *transparent* half of this front — re-competed, vehicle-routed, and datable on the public record — in sharp contrast to the opaque, export-gated, sovereign procurement of the catcher half. At that field-collection end, procurement is frequently under classification or NDA and surfaces only via named-outlet reporting or occasional procurement-portal notices. Specific catcher-vendor sovereign deal values are not publicly isolable and are carried as channel structure only, never as a sized figure.

3.3 The Contenders

Tactical intelligence is the front where the candidate list is shortest before you even start grading, and the reason is structural rather than accidental. The most-covered, most-named cohort in the popular surveillance discourse – the remote mobile-implant vendors – sits *outside* the boundary this front is built on. This front is field-deployable RF and proximity collection against a present target (IMSI/IMEI catchers, cellular interception, Wi-Fi/Bluetooth collection, direction-finding and geolocation) plus the adjacent device-in-hand modality of edge forensics, defined by *physical proximity and operational tempo*. A remote software implant delivered over the network is a different modality entirely, carved out of graded contention and treated only as market-force commentary in the Plays section. Remove that cohort and re-source what remains to dated public pages with verbatim positioning pillars, and the honest gradeable set is small.

Seven vendors are graded, not eight – and the boundary is still the finding. The cap for this report is eight contenders per front. This front comes in under the cap *by design*, and this is close to the inverse of the Fusion front. There, six graded under a deep candidate list because the boundary test (sell the analytical layer standalone) cut otherwise-rich vendors on architecture-fit. Here, seven grade because the gradeable universe is genuinely thin once two forces are applied. First, the never-graded mobile-implant carve-out removes the most-covered cohort from contention. Second, there is real public-material scarcity among the smaller RF specialists – Mer Group / Athena GS3, Wintego, and Cellxion at the margin lack the dated self-positioning that grading requires.

The count moved from six to seven on new public evidence, not a relaxed bar. Cognyte, previously held out of this front for want of a fitting public page, is graded here after its own dated border-protection page and a dated tactical-SIGINT deal surfaced as gradeable material for this front. The re-grade is captured in the report's decision record superseding the original force-grade guard. Stating that the honest count is seven, on dated public pillars, is more useful than manufacturing the eighth grade by relaxing the sourcing bar. The seven are placed across three tiers. They are 3 Gravity (the device-forensics anchors a buyer cannot map the field-collection market without naming), 3 Attention (the tactical-SIGINT specialists shaping 2026 RF-collection demand), and 1 Wildcard (a marketing-thin cellular-interception vendor graded at the edge of the rubric, and held there with explicit caution). The candidates that did not make the cut, and why, are documented at the end of this section so the boundary can be defended.



The seven graded contenders across collection modality (air-interface RF intercept ◀—▶ device-in-hand forensics) and mission orientation. Vendor positions are conceptual, not data-derived. Remote mobile-implant exploitation (NSO/Candiru/Intellexa class) is a carved-out modality, not depicted; Cognyte is graded here in the Attention tier on its dated border-protection sensor-fusion page and a dated tactical-SIGINT deal. Not investment advice.

Gravity tier

Cellebrite — the primary

⚠ Cellebrite (NASDAQ: CLBT) is a public issuer; the trading-position disclosure in the front-matter applies. This grade is built solely from public material.

"The Industry Standard for Lawfully Accessing and Collecting Digital Data ... Access critical digital evidence lawfully from feature and smartphones, drones, SIM cards, SD cards, GPS devices, and more ... Extract data from anywhere, whether in the lab, a remote location, or in the field." — Cellebrite, UFED product page, accessed 2026-06-14⁴³

Cellebrite is the category-defining device-extraction standard. It is the *de facto* reference against which every other forensics vendor in this front is positioned, and the first name a buyer cannot map the field and edge-collection market without naming. Its UFED line is marketed as the lawful-collection standard,

and the stated wedge is the broadest device and operating-system access coverage, extended past locks and encryption to the full file system. The public page tells buyers not to “let complicated locks, encryption barriers, deleted and unknown content prevent critical evidence from coming to light,” and markets “access to containerized and encrypted data to uncover every device’s Full File System (FFS) and protected data.”⁴³ The 2026 R&D narrative runs on two axes — access coverage and field deployment. The Spring 2026 Release publicly claims it “expands device access capabilities across the widest range of iOS and Android devices and operating systems, including support for iPhone 17 and iOS 26,” and “strengthens AI-powered decoding.”⁴⁴ On the field axis, the UFED variant is marketed to “extract data from anywhere, whether in the lab, a remote location, or in the field,”⁴³ which is precisely the proximity-and-tempo modality this front is defined by.

The target buyer is law enforcement, defense, national security, and government investigators, with corporate and enterprise investigations as a secondary base. On pricing, the public record is a subscription/ARR model disclosed at the aggregate level as a public issuer; there is no clean per-seat public figure. Architecturally, Cellebrite markets a dedicated-collection hardware plus extraction-software stack across the logical, full-file-system, and physical extraction ladder, with AI-decoding positioned as the modernization layer. I carry that as positioning, not verified capability, because true internal access against any specific locked device cannot be read from outside. Sourcing here is the richest in the front — vendor self-positioning, heavy named-outlet coverage, and SEC-filer disclosure. Cellebrite reported Q2 2026 revenue of USD 131.1M (up 16% year-over-year), ARR of USD 507.8M (up 21%), and GAAP net income of USD 6.4M, per its 2026-08-13 release. The reported ARR is below the USD 510M–513M Q2 range the company itself guided on 2026-05-14, and on the same date the company lowered its full-year 2026 ARR outlook to USD 550M–560M while raising its adjusted-EBITDA target. Q1 2026 was revenue USD 128.3M and ARR USD 493M.⁴⁵ Its public financial disclosures make it one of the front’s clearest war-chest anchors, alongside MSAB — a comparison the War Chests section carries.

MSAB

“The trend in 2026 is moving extraction to the field ... XRY Frontline (Kiosk, Tablet, Express) [deployed] at police stations and borders ... MSAB’s latest innovation allows agencies to utilize high-power GPUs to accelerate the brute-force process. What used to take weeks can now be accomplished in hours.” — MSAB, “Mobile Forensics in 2026,” dated 28 April 2026, accessed 2026-06-14⁴⁷

MSAB is the field-first forensics anchor — the vendor whose 2026 public narrative maps most cleanly onto this front’s proximity-and-tempo boundary. It positions itself as a “Trusted Partner in Digital Forensics” built around the XRY extraction and XAMN analysis lines.⁴⁶ Its 2026 guidance foregrounds the field shift directly: pushing the extraction act out of the central lab and to the point of seizure through “XRY Frontline (Kiosk, Tablet, Express)” form factors deployed “at police stations and borders.”⁴⁷ Where Cellebrite leads on access-coverage breadth, MSAB’s stated wedge is *decentralizing the extraction*. Its public answer to the encryption headwind is compute, framed in its own words as using “high-power GPUs to accelerate the brute-force process” so that what “used to take weeks can now be accomplished in hours.”⁴⁷ The target buyer is law enforcement, border and customs authorities, defense, and government, with a base oriented to operational field units rather than only central labs.

On pricing, MSAB is Stockholm-listed (Micro Systemation AB), so economics are disclosed at the financial-statement level; there is no clean per-seat public figure. Architecturally it markets the logical, full-file-system, and physical extraction ladder, with the Frontline field form factors and GPU brute-force as the 2026 modernization layers. A drone-forensics extension (XRY Drone) is added as a form-factor widening, carried via aggregator commentary.^{47 63} On freshness, MSAB's year-end 2025 report shows full-year net sales of SEK 462M (up 14.1%; up 19.7% currency-adjusted). Operating profit was SEK 68M at a 14.7% margin, with Q4 net sales of SEK 140.6M (up 19.6%).⁴⁸ I read MSAB and Cellebrite together as the two device-forensics anchors converging on the field edge of this front from opposite starting points. Cellebrite annexes the field from access-coverage leadership; MSAB builds the field-first posture as its core identity. The Plays section tracks this as the front's lab-to-field play.

Magnet Forensics

"Bring together evidence recovery, analysis, and insight ... Recover, analyze and report on data from mobile, computer, cloud, and vehicle sources ... Bring computer, mobile, cloud, vehicle, and acquired data into a single Axiom case." — Magnet Forensics, Magnet Axiom product page, accessed 2026-06-14⁴⁹

Magnet Forensics is the analysis-led forensics anchor — the third device-forensics name a map of this front cannot omit, and the one whose ownership change is itself a market-structure datapoint. Its AXIOM platform is marketed as multi-source evidence consolidation: pulling mobile, computer, cloud, and vehicle data into one case file with "industry-leading artifact coverage,"⁴⁹ positioned as the analyst-side complement to raw extraction. Where Cellebrite leads on getting *into* the device and MSAB on getting extraction *to the field*, Magnet's stated wedge is *unifying and analyzing* the recovered evidence across sources, promising to "access the most data possible, see the full story, and reduce time to evidence."⁴⁹ The target buyer is law enforcement, government, defense, and enterprise investigators, supported by a large practitioner community (the Magnet User Summit).⁵⁰

The ownership fact is central and frequently elided. Magnet was taken private by Thoma Bravo in a transaction valued at roughly USD 1.3–1.35B (announced January 2023; plan-of-arrangement closed April 2023) and brought together with Grayshift; Grayshift had announced a strategic growth investment from Thoma Bravo in June 2022 — into a single end-to-end digital-investigations entity.⁵¹ On pricing, that take-private removed the prior public disclosure: Magnet is now private, with no disclosed per-seat or aggregate ARR figure. Architecturally it markets a multi-source ingest, artifact-parsing, and analytics stack (AXIOM), now paired with Grayshift's GrayKey mobile-access capability under the combined entity. I read the Grayshift combination as the inorganic route to an end-to-end "unlock then extract then analyze" workflow. It buys the access layer to pair with analysis-led incumbency, against Cellebrite's organically-built breadth, and the Plays section tracks it as the front's private-equity consolidation play. The combination also removed a public issuer from the front's disclosure surface, with Cellebrite and MSAB remaining public-financial references.

Attention tier

Rayzone Group

"Full-scale, last-mile SIGINT capabilities ... Tactical RF interception for 2G, 3G, 4G, and 5G networks ... Designed for stationary, portable, and airborne deployment ... Tactical awareness. Real-time control ... Real-Time Situational Awareness at the Tactical Edge." — Rayzone Group, Proximity-Data Intelligence page, re-fetched 2026-06-14⁵²

Rayzone Group is the clearest public exemplar of the tactical "last-mile SIGINT" specialist — this front's boundary stated almost word-for-word in the vendor's own lexicon. Tel Aviv-headquartered and private, Rayzone markets its Proximity-Data Intelligence line as "full-scale, last-mile SIGINT capabilities" delivering "tactical RF interception for 2G, 3G, 4G, and 5G networks," explicitly "designed for stationary, portable, and airborne deployment."⁵² Its public range spans the catcher, the counter-catcher, and geolocation. Publicly, the Piranha unit is described as a software-defined-radio IMSI-catcher collecting mobile-phone identifiers across 2G–5G environments, and it markets a counter-product, ArrowCell, for IMSI-catcher detection and elimination.⁵³ The stated wedge is a one-vendor tactical RF stack — collection, counter-collection, and proximity intelligence — pitched to national agencies and tactical teams.

The target buyer is national agencies, internal-security services, and tactical units. Named-outlet profiles describe a client base concentrated in Asia, Central America, and Europe, which I carry as named-outlet-attributed color rather than asserted fact.⁵⁴ On pricing, Rayzone is private and undisclosed; funding and ownership circulate via aggregator listings and are carried as undisclosed, not as facts.⁵⁴ Architecturally it markets software-defined-radio collection across 2G–5G in modular, multi-form-factor configurations. Consistent with the caution that governs every RF specialist in this front, "covers 2G–5G" is a software-breadth *positioning* claim, not a verifiable statement of live 5G stand-alone content interception. Sourcing rests on the vendor's own current pages for the product positioning — the live Proximity-Data Intelligence page was re-confirmed 2026-06-14 carrying the pillars verbatim⁵² — while the geographic-client and funding color is named-outlet-attributed; no public 2026 financial disclosure exists, as Rayzone is private.

Septier Communication

"Capturing up to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi ... cellular identities, location, and other information from GSM, UMTS, LTE, and 5G devices in its coverage area ... available in various configurations to meet a range of tactical requirements." — Septier Communication, Septier Guardian product page, re-confirmed 2026-06-14⁵⁵

Septier Communication is the breadth-of-capture tactical specialist — the second Attention-tier IMSI-catcher vendor, Israel-headquartered and private, whose public wedge is the number of identifiers it captures and the range of form factors it ships. Its Guardian IMSI Catcher is marketed as "capturing up to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi," extracting

“cellular identities, location, and other information from GSM, UMTS, LTE, and 5G devices in its coverage area.”⁵⁵ It is also marketed as “available in various configurations to meet a range of tactical requirements,” including a UAV-mounted airborne variant.⁵⁶ Septier pairs the catcher with a direction-finding line, and its Cellular Locator is marketed as “a tactical location tracking solution that provides accurate location information for cellular devices, using advanced direction finders such as Septier Hunter.”⁵⁵ Where Rayzone foregrounds the SDR catcher-and-counter-catcher stack, Septier’s stated differentiation is breadth of identifier capture plus an integrated catcher-locator-direction-finding platform — a full tactical cellular-intelligence platform rather than a single-function box.

The target buyer, named on Septier’s own audience pages, is intelligence agencies, police, and counter-terrorism units.⁵⁶ On pricing, Septier is private and undisclosed. Architecturally it markets an active/passive cellular catcher across 2G–5G plus Wi-Fi and Bluetooth identifier capture and direction-finding (Hunter), in ground, portable, and airborne form factors. The same caution as Rayzone applies: the “11 identifiers” count and the 5G support are positioning sourced to the vendor’s own pages, not externally verified capability. Sourcing rests on the vendor’s own pages: the Guardian, Guardian Airborne, and Cellular Locator pages were re-confirmed live 2026-06-14, clearing the earlier sourcing flag on the exact identifier-count pillar.⁵⁵ ⁵⁶ No public 2026 financial disclosure exists (private).

Cognyte

Banner: Cognyte is graded here on dated public material only, with no internal knowledge used. The author served as VP Product at Cognyte; this grade is independent ProductBeacon research built from Cognyte’s own current product page and a dated public deal announcement, and reflects the author’s personal analysis, not Cognyte’s position. The gap between Cognyte’s soft public headline and its own deeper public record is shown as a sourced contrast, never as an asserted intent.

*“Cognyte’s Border Protection solution takes a multi-layered approach by analyzing existing border sensors (cameras, radar, satellites and more) and fusing together with SIGINT, OSINT and VISINT sources.” — Cognyte, Border Protection / Maritime Intelligence page, cognyte.com/border-protection-maritime-intelligence-na/ (accessed 2026-06-16)*⁵⁸

Cognyte is graded in the Attention tier, and like its placements elsewhere in this report the grade rests on a distinct dated public pillar — here the field-and-border collection page, not the network-intelligence glossary that anchors the National Agencies front or the NEXYTE “data source” framing that anchors the Law Enforcement Monitoring Centers front. The primary pillar is Cognyte’s own border-protection page, which describes a solution that fuses “existing border sensors (cameras, radar, satellites and more)” together with “SIGINT, OSINT and VISINT sources.”⁵⁸ That is the proximity-and-tempo shape of this front stated in Cognyte’s own words: sensor fusion against a present operational picture at the border and checkpoint, the field-collection edge rather than the carrier core. The corroboration is a dated deal: a Tier-1 military-intelligence agency contract of roughly USD 5M, publicly described as deploying “a tactical SIGINT solution for operational missions” (BusinessWire/StockTitan, 2025-11-19).⁵⁹ The VANTAGE trademark — “telecommunication surveillance software for the integrated interception, collection, monitoring, analyzing, searching, tracking and reporting of data and voice communications,” the subject

of a USPTO-recorded 2021 assignment from Verint Systems Inc. to Cognyte Software Ltd. — corroborates the interception lineage but is not this front's pillar; it is weighted to the National Agencies front and carried as corroboration only.⁶⁰

The grade construction is explicit about what carries it. Cognyte is placed in Attention, not Gravity, because the Gravity tier here is the device-in-hand forensics anchors a buyer cannot map the field-collection market without naming. Cognyte's public footprint on this front is the field-and-border sensor-fusion line plus the dated tactical-SIGINT deal, not a standalone field-forensics product surface. An honesty note in the same register that governs the whole chapter: some tactical product names in this category are classified or non-public. So the tactical-SIGINT positioning and the dated deal are the gradeable public material, and the grade is built strictly from them — the classification of specific product names narrows what can be quoted rather than removing the grade.

The target buyer, on the border-protection page's own framing, is the border, defense, and national-security operator. On pricing, Cognyte's tactical line is not separately disclosed; its public financials are group-level (the Convergence chapter carries those). The current border-protection product positioning was directly fetched and quoted verbatim; the dated deal rests on a named-outlet release. The VANTAGE goods-description wording is **confirmed** verbatim against the USPTO record. Consistent with the front-wide discipline, none of Cognyte's lineage artifacts is asserted here as a present tactical *capability* — they are dated records of positioning and a dated deal, quoted and attributed. Cognyte is a cross-front vendor. It is also graded in the National Agencies front (the network-intelligence glossary), the Law Enforcement Monitoring Centers front (the NEXYTE lawful-interception "data source" framing), the Fusion / Decision Intelligence front (the NEXYTE decision-intelligence platform), and Open-Source Intelligence (LUMINAR WEBINT). The pillar graded here on this front — the border-protection "SIGINT, OSINT and VISINT" line — is distinct from each of those and is logged for the cross-front reconciliation.

Wildcard tier

Cellxion

Wildcard caution: Cellxion's own current self-positioning page is thin. The pillar below is drawn from named-outlet and surveillance-industry-index descriptions, dated and explicitly attributed as third-party characterization rather than vendor self-positioning. The read stays descriptive throughout; this is a borderline grade that may be demoted to the Plays-section market-force discussion if a live dated vendor pillar cannot be captured.

Cellxion is publicly described — via the Surveillance Industry Index and named directory listings — as a UK-based supplier of cellular-interception and IMSI-catcher equipment (the GX/UGX cellular-intercept range), positioned for covert tactical cellular monitoring. — surveillance-industry-index / named-listing description, accessed 2026-06-14⁵⁷

Cellxion is the Wildcard, and its Wildcard status is precisely its thinness rather than its trajectory. It occupies the specialist cellular-interception niche — a smaller, lower-profile supplier in the same IMSI-catcher market as Rayzone and Septier. A real public footprint exists: it appears in surveillance-industry indices and directory listings as a cellular-interception vendor with the GX/UGX range.⁵⁷ Yet it lacks the rich self-positioning of the Attention-tier specialists, so its continued presence in a niche being

structurally compressed by encryption (per the headwind on classic passive content interception) is a descriptive market-force question, not an assertive read. I do not characterize its prospects in either direction.

The target buyer, per the index and listing descriptions, is law enforcement and security and intelligence agencies.⁵⁷ On pricing, Cellxion is private and undisclosed. Architecturally, the classification is cellular-interception / IMSI-catcher hardware (the GX/UGX range, per the index listing), carried descriptively as named-outlet and index characterization; no capability claim is asserted as fact. Sourcing is the weakest of the graded set — named-outlet and surveillance-industry-index description, explicitly not vendor self-positioning — and the grade is held strictly to the report's Wildcard caution: descriptive only, no pattern claim, no negative claim. No 2026 vendor disclosure surfaced. Cellxion is carried as a continuing niche presence on the strength of index listings, with the thinness stated as the finding.

The cut line — who is not graded here, and why

The front's apparent depth in the popular discourse forces a defensible cut. On this front the cut is driven by three forces: the never-graded mobile-implant carve-out, genuine public-material scarcity among the smaller RF vendors, and the boundary that excludes diversified primes. Each candidate below is real; each is placed where the public record actually puts it.

- **Cognyte IS graded on this front — in the Attention tier above.** An earlier pass concluded that no dated public cognyte.com page fit this front's proximity-and-tempo boundary, and held Cognyte out of this front. On a fuller reading of Cognyte's *own* dated public record, that conclusion was too narrow. Cognyte's border-protection page describes a field-and-border collection solution fusing "existing border sensors (cameras, radar, satellites and more)" with "SIGINT, OSINT and VISINT sources,"⁵⁸ and a dated public release describes a Tier-1 military-intelligence deal deploying "a tactical SIGINT solution for operational missions" (2025-11-19).⁵⁹ Those carry the grade on this front, in the Attention tier (the re-grade supersedes the original force-grade guard, captured in the report's decision record). The true residual remains stated, not papered over: some tactical product names in this category are classified or non-public. But the gradeable public material is the border-protection page and the dated deal, and the grade is built strictly from them, with no internal knowledge of any Cognyte tactical product line used anywhere in this chapter.
- **Mer Group / Athena GS3** — no current dated standalone tactical-collection self-positioning that fits this front's boundary surfaces. Mer is a diversified communications and security group, and Athena GS3's public footprint is thin and dated. Referenced in the Plays section as market color; the thinness is stated, not papered over.
- **Wintego** — described historically by named outlets (the WINT Wi-Fi man-in-the-middle "backpack" system, 2016) but with no current dated self-positioning page that the grading test requires.⁶² Carried in the Plays section as descriptive modality context only; not graded, and no operating-status claim is made.
- **Rohde & Schwarz / L3Harris / PKI Electronic Intelligence** — these appear in the IMSI-catcher market sizing as named majors,⁶¹ but they are diversified RF and defense primes whose tactical-collection products are a small slice of a much larger business. On this front, a prime's business line is gradeable only where its own dated public product-line page supports proximity RF collection against a present target; any financial scale is kept to the relevant disclosed business line or

segment and never inferred from total-parent revenue. These three remain market references because that front-local product-line and sizing record does not support a primary specialist grade here.

- **The locked never-graded set – NSO, Candiru, Intellexa, Ability, Voyager Labs – is never graded.** Remote mobile-implant exploitation (the Pegasus / Predator class) is a technically distinct modality – a remote software implant delivered over the network, governed by different export-control language (intrusion-software controls) than the RF-proximity hardware this front grades. It is carved out of graded contention by construction and appears only as descriptive market-force commentary in the Plays section (the encryption-and-accountability play), where any sanctions or distress statement is attributed to a named outlet or official record. No grade, no pattern claim, no negative-intent framing attaches to any vendor in this cohort.

A closing note on what the boundary buys. One discipline holds the graded set at seven: re-source to a dated public page with a verbatim pillar, carve out the modality that does not fit, and grade a vendor only where its footprint is publicly readable. It is the same discipline that keeps every placement above honest, and it moved Cognyte into the graded set once its own dated public pages supported it. Grade the field-collection layer on public material. Reference the rest.

3.4 Their Plays

Three strategic moves are in motion across the Tactical Intelligence front in 2026. The front carries a deliberately small play count – three, not five – because its evidence floor is the strictest in the report: every play here is anchored to a dated public event, and a play without a public datable anchor is dropped, not softened. Each names its participants, signals its opinion explicitly, and carries a falsifiable read grounded on a publicly observable proxy. The shape of the three is the shape of the front itself. One is the modality migration that is annexing the field-collection edge to the device-forensics cohort. One is the private-equity consolidation reshaping who owns the forensics stack. One is the twin squeeze – encryption from below and export-control accountability from above – bearing down on the classic RF-catcher modality and the broader surveillance market it sits next to. The carved-out locked never-graded set (NSO/Candiru/Intellexa class) appears in the third play only as descriptive, named-outlet-attributed market-force commentary – never graded, never pattern-claimed, never assigned intent.

Play 1: The Lab-to-Field Shift – Extraction Moves to the Point of Seizure

- **Observation.** The device-forensics anchors are publicly relocating the act of extraction out of the central lab and to where the device is seized. MSAB states it plainly: “The trend in 2026 is moving extraction to the field,” answered by “XRY Frontline (Kiosk, Tablet, Express)” deployed “at police stations and borders” (MSAB, dated 28 April 2026).⁶⁴ Cellebrite markets UFED extraction to “Extract data from anywhere, whether in the lab, a remote location, or in the field,”⁶⁵ and its Q1 FY2026 portfolio commentary reports a drone-forensics expansion (reported 2026-05-14).⁶⁶ Both vendors field ruggedized and field-form-factor hardware, and the encryption answer is being reframed as compute – MSAB markets that agencies can “utilize high-power GPUs to accelerate the brute-force process,” so that “what used to take weeks can now be accomplished in hours.”⁶⁴
- **Participants.** MSAB, Cellebrite, Magnet, and the deployed government field units – border, customs, arrest, and checkpoint teams – that are the named deployment surface in the vendors’ own pages.

- **My read.** I read this as the forensics cohort annexing the field-collection edge of this front, not merely modernizing the lab. Pushing the extraction act to the point of seizure maps directly onto this front's defining property — physical proximity plus operational tempo. And the device-in-hand modality, where brute-forcing an unlock sidesteps the transport-layer encryption that compresses the over-the-air RF-catcher modality, is the structurally healthiest part of the front. The forensics anchors are walking into the tactical edge from a position of growth, while the classic catcher modality is walking out of it under encryption pressure (Play 3).
- **Falsifiable read.** *If through H2 2026 the named forensics vendors publish further field and edge form factors — kiosks, drones, ruggedized units — and name government field-deployment references, the lab-to-field thesis hardens into the front's defining migration. If instead their 2026 roadmaps and named wins re-center on lab-based throughput and back away from field form factors, the convergence read weakens and edge-forensics stays a niche of the lab market rather than the new tactical center of gravity.*
- **Evidence.** MSAB "Mobile Forensics in 2026" guidance, 28 April 2026⁶⁴; Cellebrite UFED field-extraction positioning⁶⁵; Cellebrite Q1 FY2026 drone-forensics expansion, reported 2026-05-14⁶⁶.

Play 2: Consolidation in Device Forensics Under Private Equity

- **Observation.** The category's defining capital event is private-equity consolidation at the forensics layer. Thoma Bravo took Magnet Forensics private: the plan of arrangement closed 2023-04-06 at CA\$44.25 per share, with aggregate cash consideration of approximately CA\$1.044B plus rollover shares of approximately CA\$0.628B. The transaction was reported at a headline value of roughly C\$1.8B (US\$~1.3B per-share equity). It then combined Magnet with Grayshift (after Grayshift's June 2022 strategic growth-investment announcement) into a single end-to-end digital-investigations entity.⁶⁷ ⁶⁸ The combined entity operates privately as of 2026, with no public re-listing or fresh disclosed valuation in the window.⁶⁷
- **Participants.** Thoma Bravo, Magnet Forensics, Grayshift, and the law-enforcement and government forensics buyer base.
- **My read.** I read this as the front's consolidation mode: buy the missing layer and assemble the full workflow under one owner. The analysis-led forensics leader (Magnet/AXIOM) and a mobile-access and unlock specialist (Grayshift/GrayKey) were merged under PE ownership into a combined access-plus-analysis stack — the inorganic route to an end-to-end "unlock then extract then analyze" workflow, set against Cellebrite's organically built breadth. The move also removed a public issuer from the market's disclosure surface, taking Magnet's financials dark and leaving Cellebrite and MSAB as public-financial references.
- **Falsifiable read.** *If H2 2026 brings a second comparable acquisition of an access, extraction, or analysis specialist by a forensics consolidator, the buy-the-stack pattern is the front's structural direction. If instead the combined Magnet/Grayshift entity shows only feature-level integration without named end-to-end government wins, organic breadth — Cellebrite's route — is out-competing the assembled stack, and the consolidation is a financing story rather than a competitive realignment.*
- **Evidence.** Thoma Bravo closing of Magnet plan of arrangement, 2023-04-06⁶⁷; transaction value reported at C\$1.8B⁶⁸; Grayshift merged into Magnet following the take-private⁶⁷.

Play 3: The Encryption Headwind and the Export-Control Overhang Compress the RF-Catcher Modality

- Observation.** Two compounding public forces squeeze the classic RF-catcher modality and the surveillance market it neighbors. First, **encryption and legacy-network sunset** structurally compress passive content-interception: the catcher's reliable yield narrows toward identity, metadata, and location rather than content. Market commentary already flags 5G-SA-compatible interception as white space "where no vendor has yet established a clearly dominant globally deployed commercial solution."⁶⁹ Second, an **accountability and export-control overhang** reshapes the broader surveillance market's structure. The US Commerce Department added NSO Group and Candiru to the Entity List on 4 November 2021 under a "presumption of denial," on a determination they "developed and supplied spyware to foreign governments" used to "maliciously target government officials, journalists, businesspeople, activists, academics, and embassy workers." It added Intellexa/Cytrox on 18 July 2023.⁷⁰ The EU Dual-Use Regulation 2021/821 Article 5 cyber-surveillance catch-all – with European Commission guidelines issued 15 October 2024 – imports a human-rights end-use test reaching RF and cellular surveillance items.⁷¹
- Participants.** The RF-catcher specialists (Rayzone, Septier, Cellxion) on the encryption side; the carved-out locked never-graded set (NSO, Candiru, Intellexa) and the US and EU regulators on the accountability side. The mobile-implant vendors are named only as a carved-out market force – a remote-software-implant modality technically distinct from this front's RF-proximity hardware – with no grade, no pattern claim, and no negative-intent framing. The sanctions facts are attributed to US official records and named outlets, dated.
- My read.** I read this as a front being squeezed from two directions at once. The classic over-the-air content-interception value proposition is eroding technically as encryption advances and 2G/3G networks sunset, pushing tactical RF vendors toward active and hybrid catchers, direction-finding and identity value, and the device-in-hand forensics route. At the same time, the broader surveillance market operates under a hardening export-control and legitimacy regime that raises the cost and reputational risk of cross-border sale. That is the same regime that, per the war-chest evidence, discourages public funding signaling among the private tactical specialists.
- Falsifiable read.** *If through H2 2026 and into 2027 tactical vendors' public pages and named-outlet coverage keep foregrounding forensics, direction-finding, identity, active and hybrid catchers, and 5G-SA research over classic passive-content interception – and export-control actions continue – the squeeze thesis holds. If instead a vendor publicly demonstrates durable 5G-SA content interception at scale, or the export-control regime visibly loosens, the read weakens.*
- Evidence.** IMSI-catcher market commentary on 5G-SA white space⁶⁹; US Commerce/BIS Entity List additions – NSO and Candiru (4 November 2021), Intellexa/Cytrox (18 July 2023)⁷⁰; EU Dual-Use Reg. 2021/821 Article 5 and the 15 October 2024 guidelines⁷¹.

3.5 War Chests & Casualties

A snapshot of the capital behind the Tactical Intelligence front – recent rounds and public status, public valuations or disclosed figures, the strategic investors and owners, and any distress signal that meets the evidentiary bar. Every figure traces to a named outlet, a filing, or the vendor's own dated release. Public-company figures (CLBT, MSAB B) are taken at the disclosed level; private-equity-owned Magnet is recorded at its disclosed take-private deal structure, with current operating financials recorded as private and undisclosed rather than estimated. Private vendors (Rayzone, Septier, Cellxion) enter only via a named-outlet figure or filing, dated and attributed; absent that, they read "private, undisclosed." No internal-CI private-vendor revenue figure is carried. The Distress Signal column is populated only where a same-paragraph public footnote anchors a specific event; absent that, it reads "none observed." Live market caps are deliberately not fixed here – they are reported at the disclaimer-governed figure on the publication date.

Vendor	Most Recent Round / Public Status (dated)	Valuation / Disclosed Figure (public only)	Strategic Investor or Owner	Distress Signal (-gated)
Cellebrite (IL/US — NASDAQ: CLBT; mobile forensics / UFED / tactical collection family)	Public (NASDAQ: CLBT). Q2 FY2026 (qtr ended 2026-06-30, reported 2026-08-13): ARR \$507.8M, +21% YoY ; total revenue \$131.1M, +16% YoY; subscription revenue \$119.5M (+16%); adj. EBITDA \$31.8M (24.2% margin); net revenue retention 117%; Q1 FY2026 (reported 2026-05-14): ARR \$493M, +21% YoY; total revenue \$128.3M, +19% YoY. FY2026 ARR outlook lowered on 2026-08-13 to \$550M–\$560M (14–16% growth) from the \$567M–\$573M issued 2026-05-14 ; FY2026 revenue outlook \$555M–\$561M; adjusted-EBITDA target raised to \$153M–\$159M. CEO: the company announced on 2026-08-13 that Shiven Ramji succeeds Thomas E. Hogan as chief executive, described in the company's own Form 6-K as a planned transition; Hogan also stepped down from the board (Form 6-K, 2026-08-13) ^{72 73 84} ,	ARR \$507.8M / Q2 revenue \$131.1M is the disclosed anchor ⁷² ; the Q1 FY2026 print (ARR \$493M / revenue \$128.3M) stays as the prior dated quarter. <i>Live market cap reported disclaimer-governed at publication date.</i>	Publicly traded; no controlling strategic owner	None observed — ARR +21%, federal pipeline +35%, FedRAMP High authorization ⁷²
MSAB (SE — Nasdaq Stockholm: MSAB B; mobile forensics / XRY / field extraction)	Public (Nasdaq Stockholm: MSAB B). FY2025 (year ended 2025-12-31): net sales SEK 462M, +14.1% YoY (currency-adjusted +19.7%); EBIT SEK 68M (14.7% margin); profit after tax SEK 49.2M; EPS SEK 2.66 ⁷⁴ . Interim H1 2026 (Q2 report, dated 2026-07-16): net turnover SEK 122.5M, +52.1% YoY; first-ever profitable first half ⁸³	FY2025 net sales SEK 462M is the disclosed anchor ⁷⁴ ; the H1 2026 interim is the freshest disclosed print ⁸³ . <i>Live market cap reported disclaimer-governed at publication date.</i>	Publicly traded; no controlling strategic owner	None observed — company calls FY2025 “the most successful year in [its] history” ⁷⁴

Vendor	Most Recent Round / Public Status (dated)	Valuation / Disclosed Figure (public only)	Strategic Investor or Owner	Distress Signal (-gated)
Magnet Forensics (CA — private; digital-investigation suite + Grayshift mobile-access, merged)	Private (Thoma Bravo portfolio). Take-private by Thoma Bravo closed 2023-04-06 ; CA\$44.25/share; aggregate cash consideration ~CA\$1.044B + rollover shares ~CA\$0.628B; headline transaction value ~C\$1.8B (US\$~1.3B per-share equity). Grayshift (June 2022 strategic growth investment; terms undisclosed) merged into Magnet at/after close ^{75 76 77}	Private, undisclosed. No public valuation or operating financials since the 2023 take-private — recorded as such, not estimated	Thoma Bravo (PE; Magnet acquirer at the 2023-04-06 close); Grayshift's 2022 investment terms undisclosed ^{75 77}	None observed — no public distress event; no disclosed financials to assess
Rayzone Group (IL — private; tactical SIGINT / proximity-data intelligence / Piranha)	Private. No disclosed funding round or priced valuation in public sources ⁷⁸ . The only dated named-outlet figure is a 2019 reported turnover of three group companies of ~NIS 107M (~US\$33M) (Calcalist/CTech, dated 2019) ⁷⁹	Private, undisclosed for any current valuation. The 2019 ~NIS 107M turnover line is the only named-outlet figure ⁷⁹ ; aggregator revenue bands are directional only and not load-bearing ⁷⁸	Privately held (Israel); no disclosed institutional lead in public sources ⁷⁸	None observed
Septier (IL — private; cellular interception / positioning / Septier Guardian family)	Private. Aggregators conflict — Tracxn records no raised funding; PitchBook lists ~\$1.6M — neither a load-bearing public figure ⁸⁰ . Company self-describes as a “fast-growing, highly profitable private company” (its own About page) ⁸¹	Private, undisclosed. No priced valuation in public sources ⁸⁰	Privately held (Israel); no disclosed institutional owner in public sources ⁸⁰	None observed
Cellxion (UK — private; IMSI-catchers / cellular interception)	Private. No disclosed funding or valuation. Public footprint is procurement-side: named-outlet reporting of a >£1M Metropolitan Police purchase (2015) and subsequent UK LEA/HMRC contracts (Motherboard/VICE, dated) ⁸²	Private, undisclosed. No priced valuation or disclosed owner in public sources ⁸²	Privately held (UK); no disclosed institutional owner in public sources ⁸²	None observed

Commentary on the capital pattern (≤150 words)

This front combines **public financial disclosure with a private long tail**. Cellebrite (CLBT) and MSAB (MSAB B) both disclose financials and are growing: Cellebrite at ARR \$507.8M, +21%⁷² dwarfs MSAB's SEK 462M FY2025 net sales⁷⁴, but both report record momentum, not distress. The category's defining capital event is **private-equity consolidation at the forensics layer**: Thoma Bravo took Magnet private in 2023 (~C\$1.8B headline) and folded Grayshift in⁷⁵,⁷⁷, removing the second-largest mobile-forensics name from public markets and ending visibility into its numbers. The Israeli tactical specialists (Rayzone, Septier) and the UK IMSI-catcher house (Cellxion) raise little disclosed venture capital and publish no priced valuations — consistent with a field-collection segment where customer secrecy and export-control exposure discourage public funding signaling⁷⁸,⁸⁰,⁸².

3.6 Winning & Losing

Three pattern claims carry this front. Each begins with a dated public observation, separates the observation from my read of it, and ends with a conditional that a publicly observable proxy can confirm or refute before the next refresh. The throughline they share is the one structural fact the whole chapter is built on. Air-interface encryption is compressing the classic over-the-air interception modality, and the demand it sheds is migrating to two places — the device in hand, and the accountable end of the supply side. The three claims trace that migration from three angles: where the growth went (device/edge forensics), where the boundary is dissolving (lab to field), and who gets to buy and sell at all (the surveillance-market reckoning).

Pattern Claim 1 – The Edge-Forensics Ascendancy

Observation. The growth on this front is not in the catcher; it is in the seized device. The classic passive over-the-air interception value proposition is structurally compressing. End-to-end encryption on WhatsApp, Signal, and Telegram, plus modern device-OS encryption, mean intercepting the *wire* increasingly yields ciphertext, and 2G/3G sunset is steadily removing the downgrade target that catchers relied on for content.⁸⁵ ⁸⁶ The public market figures show the split directly. The cellular-interception / IMSI-catcher segment runs at a flat-to-modest CAGR – Mordor Intelligence brackets the IMSI-catcher market at USD 207.68M (2026) growing at an 8.08% CAGR, and the broader cellular-interception framing at a 4.61% CAGR through 2031. The mobile-forensics market that the device-in-hand modality belongs to runs materially faster, with MarketsandMarkets bracketing it at USD 5.72B (2026) at an 11.8% CAGR and Mordor placing the digital-forensics parent at a 16.5% CAGR on the edition originally cited, 12.68% (2026–2031) on its 2026-07-23 re-issue.⁸⁷ ⁸⁸ The single hardest datapoint anchors the same conclusion. Cellebrite (NASDAQ: CLBT), a public issuer in this front, reported Q2 FY2026 ARR of USD 507.8M, up 21% year-over-year, at 80.8% GAAP gross margin and 117% net revenue retention (Q1 FY2026: USD 493M, +21%), while lowering its full-year ARR outlook to 14–16% growth. MSAB (Nasdaq Stockholm: MSAB B) reported FY2025 net sales of SEK 462M, up 14.1%, which the company called “the most successful year in the company’s history.”⁸⁹ ⁹⁰

My read. I read this as the defining structural fact of the front. When the device is in hand and the unlock can be brute-forced, the operator sidesteps the transport-layer crypto that defeats the over-the-air catcher entirely. So the structurally healthy growth sits in device and edge forensics (Cellebrite, MSAB, Magnet), and the RF-catcher modality is the laggard, kept alive mainly by the identity-and-location use case that does not require breaking content encryption. The catcher is not dead; its yield is just narrowing from *content* to *presence*, while the dollars accrete on the forensics side. The two public issuers are the proof the demand is real, recurring, and accelerating – not a publisher’s TAM derivative but disclosed, audited ARR.

Conditional prediction. If, through H2 2026 and into 2027, the device-extraction issuers keep posting low-to-mid-teens or better disclosed growth while the cellular-interception/catcher market figures stay in the mid-single-to-high-single-digit CAGR band, the edge-forensics-ascendancy read holds. If instead Cellebrite’s ARR growth decelerates below roughly 12% across two consecutive quarters, or a methodology-disclosed publisher shows the cellular-interception CAGR re-accelerating above the forensics CAGR, the read weakens – the compression of the wire would be reversing, not deepening.

Sources. Encryption-compression framing: Oxygen Forensics 2025 + EPRS PE 775879 (2025).⁸⁵ ⁸⁶ Market CAGRs: Mordor Intelligence (IMSI-catcher; cellular interception), MarketsandMarkets (mobile forensics), Mordor (digital forensics).⁸⁷ ⁸⁸ Disclosed financials: Cellebrite Q2 FY2026 (reported 2026-08-13), with Q1 FY2026 (reported 2026-05-14) as the prior dated quarter; MSAB FY2025 year-end report.⁸⁹ ⁹⁰

Pattern Claim 1 — The Edge-Forensics Ascendancy

GOVERNMENT AND LAW ENFORCEMENT INTELLIGENCE MARKETS 2026 · FRONT 3: TACTICAL INTELLIGENCE

The growth is not in the catcher — it is in the seized device. Encryption sheds demand; forensics absorbs it.

THE HEADWIND — AIR-INTERFACE ENCRYPTION

5G transport crypto + SUPI/SUCI

over-the-air content interception yields ciphertext;
the permanent identifier is encrypted

2G / 3G sunset

removes the downgrade target the catcher relied on
for content — yield narrows to identity / presence

PASSIVE RF-INTERCEPT MODALITY — COMPRESSED

IMSI-catcher market: USD 207.68M (2026), 8.08% CAGR

Cellular interception framing: 4.61% CAGR through 2031

Flat-to-modest growth — the laggard of the front

value

migrates →

THE GROWTH — DEVICE / EDGE FORENSICS

A seized device sidesteps wire encryption —
the unlock can be brute-forced

Cellebrite (NASDAQ: CLBT)

ARR USD 507.8M · +21% YoY

80.8% GAAP gross margin · 117% net revenue retention
Q2 FY2026, disclosed/audited — not a TAM derivative

MSAB (Nasdaq Stockholm: MSAB B)

FY2025 net sales SEK 462M · +14.1%

"the most successful year in the company's history"

Mobile forensics 11.8% CAGR · digital forensics 12.68%
(2026–2031 re-issued edition) — both far above the catcher band

FALSIFIABLE TEST

HOLDS IF — through H2 2026 into 2027 — the device-extraction issuers keep posting low-to-mid-teens-or-better
disclosed growth while the cellular-interception / catcher market figures stay in the mid-to-high-single-digit CAGR band.

WEAKENS IF — Cellebrite's ARR growth decelerates below roughly 12% across two consecutive quarters, or a
methodology-disclosed publisher shows the cellular-interception CAGR re-accelerating above the forensics CAGR.

Proxy: Cellebrite SEC 6-K / IR releases · MSAB interim & year-end reports · Mordor / MarketsandMarkets methodology-disclosed updates.
A reopening of the wire would reverse, not deepen, the compression.

Author's read of public material, June 2026.

Conceptual, not data-derived. Not investment advice.

Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

Pattern Claim 2 – The Lab-to-Field Collapse

Observation. Extraction is moving out of the central lab and into the field, collapsing the tactical/forensic boundary the market used to keep. MSAB states plainly in its 2026 guidance that “The trend in 2026 is moving extraction to the field,” answered by “XRY Frontline (Kiosk, Tablet, Express)” deployed “at police stations and borders,” and it unveiled XRY Drone in November 2025.^{91 92} Cellebrite markets its UFED to “Extract data from anywhere, whether in the lab, a remote location, or in the field,” delivered on a “UFED Touch3 Ruggedized Tablet,” and expanded into drone forensics in its Q1 2026 portfolio.^{93 89} The demand surface is the border and the arrest. U.S. Customs and Border Protection reported searching roughly 47,000 electronic devices in 2024, up from about 8,500 in 2015 – a ~5.5× rise over a decade. The procurement tracks the volume, with DHS/ICE planning a five-year IDIQ at a USD 100M ceiling and the FBI/DOJ awarding Cellebrite a USD 5.7M extraction contract through the NASA SEWP vehicle (start 2025-09-30).^{94 95}

My read. I read this as the device-forensics anchors annexing the field-collection edge of this front – not merely modernizing the lab. The same deployed unit that used to choose between a catcher (device on the air) and a lab extraction (device in hand) increasingly wants both at the point of seizure, in operational time. That is exactly the *proximity-plus-tempo* property that defines this front. The winner of that collapse is not whoever extracts fastest; it is whoever makes field-grade extraction *defensible* – chain-of-custody, forensic soundness, oversight – because a kiosk at a border crossing is a legal and reputational surface in a way a lab bench is not. Cellebrite’s “lawfully accessing” framing and MSAB’s forensic-soundness language are positioning for exactly that defensibility contest.

Conditional prediction. If, through H2 2026, the named forensics vendors publish further field/edge form factors (kiosks, ruggedized units, drone-forensics lines) *and* name government field-deployment references, the lab-to-field-collapse read hardens into the front’s structural direction. The read weakens on either of two signals: 2026 roadmaps and named wins re-centering on lab-based throughput and backing away from field form factors, or a named legal or policy constraint cutting the border device-search volumes that anchor the demand. On either, the convergence read weakens and edge extraction stays a niche of the lab market.

Sources. Field-shift positioning: MSAB 2026 guidance (dated 28 April 2026); XRY Drone (Nov 2025); Cellebrite UFED page + Q1 2026 portfolio.^{91 92 93 89} Demand and procurement: CBP device-search counts and DHS/ICE IDIQ via FedScoop; FBI/DOJ NASA-SEWP award.^{94 95}

Pattern Claim 2 — The Lab-to-Field Collapse

GOVERNMENT AND LAW ENFORCEMENT INTELLIGENCE MARKETS 2026 · FRONT 3: TACTICAL INTELLIGENCE

Extraction moves to the point of seizure — the tactical/forensic boundary the market used to keep is dissolving



THE DEMAND SURFACE — BORDER & ARREST

US CBP electronic-device searches



The procurement tracks the volume

- DHS/ICE planned 5-yr IDIQ, USD 100M ceiling (Cellebrite)
- FBI/DOJ awarded Cellebrite USD 5.7M extraction contract via NASA SEWP (start 2025-09-30)

The winner is not whoever extracts fastest —
it is whoever makes field extraction **defensible**
(chain-of-custody, forensic soundness, oversight).

FALSIFIABLE TEST

HOLDS IF — through H2 2026 the forensics anchors publish further field/edge form factors (kiosks, ruggedized units, drone-forensics lines) AND name government field-deployment references.

WEAKENS IF — 2026 roadmaps re-center on lab throughput with field form factors quietly retired, or a named legal/policy constraint cuts the border device-search volumes that anchor the demand.

Author's read of public material, June 2026.
Conceptual, not data-derived. Not investment advice.

Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

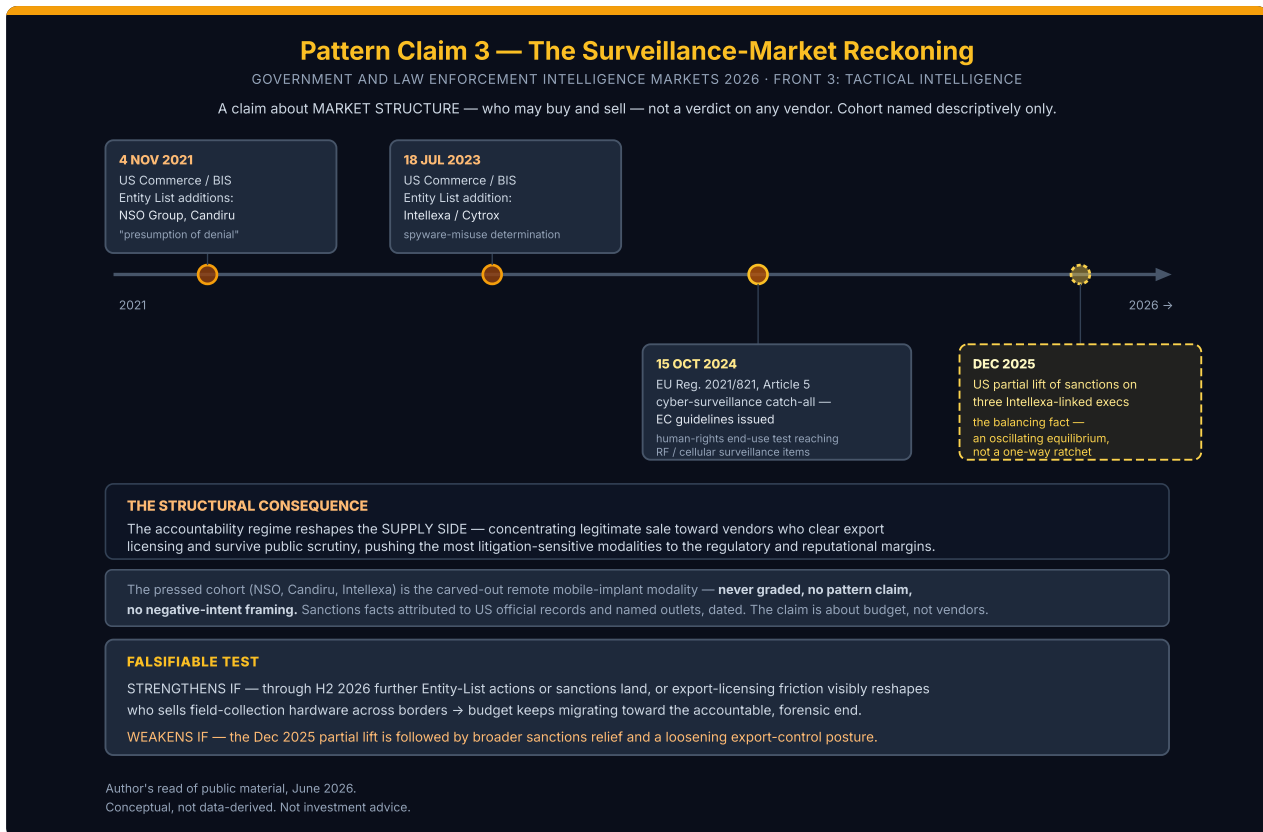
Pattern Claim 3 – The Surveillance-Market Reckoning

Observation. The structure of who may buy and who may sell in this market is being reshaped by sanctions and export-control accountability — and the cohort it presses hardest is the one this report never grades. The U.S. Commerce Department added NSO Group and Candiru to the Entity List in November 2021 under a “presumption of denial,” and added Intellexa/Cytrox in July 2023, each on determinations tied to spyware misuse.⁹⁶ The EU Dual-Use Regulation 2021/821 added an Article 5 catch-all reaching “cyber-surveillance items” — including, per EU guidance, “equipment for covertly extracting communication content and metadata that is transmitted over the air via wireless communication, and radiofrequency monitoring equipment” — importing a human-rights end-use test, with guidelines issued 15 October 2024.⁹⁷ The regime is not monotonic. In December 2025 the U.S. administration partially lifted sanctions on three Intellexa-linked executives, an action industry read as an easing for surveillance providers.⁹⁸

My read. I read this as a market-structure reckoning rather than a verdict on any one vendor. The remote locked never-graded set — NSO, Candiru, Intellexa, and the rest of the Pegasus/Predator class — is a technically distinct modality from this front’s RF-proximity hardware. It is governed by different export-control language: intrusion software under Wassenaar Category 4 / U.S. EAR 4D004, versus IP-network and RF surveillance under Category 5 Part 1 / 5A001.j. It is carved out of grading entirely and named here only as a market force, descriptively and on the official and named-outlet record.⁹⁷ What the accountability regime does, structurally, is reshape the *supply side*. It concentrates legitimate sale toward vendors who can clear export licensing and survive public scrutiny, and pushes the most litigation-sensitive modalities toward the regulatory and reputational margins. The December 2025 partial lift is the reminder that this is a contested, oscillating equilibrium, not a one-way ratchet — which is precisely why it is a market force to watch rather than a settled outcome.

Conditional prediction. If, through H2 2026, further Entity-List actions or sanctions land on surveillance-cohort entities, or export-licensing friction visibly reshapes who sells field-collection hardware across borders, the reckoning read strengthens and budget continues migrating toward the accountable, forensic end of the market. If instead the December 2025 partial lift is followed by broader sanctions relief and a loosening export-control posture, the read weakens — the legitimacy pressure that has favored the accountable vendors would be relaxing.

Sources. Entity-List actions: BIS / Federal Register (NSO + Candiru, Nov 2021; Intellexa/Cytrox, Jul 2023) via official records and named outlets.⁹⁶ EU Article 5 cyber-surveillance catch-all and the “over the air... radiofrequency monitoring equipment” definition: Akin Gump / SIPRI on Regulation 2021/821 (guidelines 15 Oct 2024).⁹⁷ December 2025 partial sanctions lift: Access Now / named-outlet reporting on the Intellexa consortium.⁹⁸



Author's read of public material, June 2026. Conceptual, not data-derived. Not investment advice.

Winners, Watch, Losers

Winners. **Cellebrite** and **MSAB** are the front's momentum winners, and the label rests on disclosed financials rather than positioning. Cellebrite reported Q2 FY2026 ARR of USD 507.8M, up 21% year-over-year, with subscription revenue up 16%, 80.8% GAAP gross margin and 117% net revenue retention (Q1 FY2026: ARR USD 493M, +21%), while lowering its full-year 2026 ARR outlook to 14–16% growth — a disclosed public-issuer scorecard in the front.⁸⁹ MSAB reported FY2025 net sales of SEK 462M, up 14.1% (currency-adjusted 19.7%), at a 14.7% EBIT margin, a record year on its own statement.⁹⁰ Both are riding the edge-forensics ascendancy of Pattern Claim 1, and both are pushing extraction to the field per Pattern Claim 2 — Cellebrite's "lab, a remote location, or in the field" UFED and MSAB's XRY Frontline at "police stations and borders."^{93 91} I read these two not as winning a share war against each other but as winning the *modality* war. The seized device is where this front's money is, and they are its public anchors. (Cellebrite is a public issuer — the trading-position disclaimer applies.)

Watch. **Rayzone**, **Septier**, and **Cellxion** are on Watch — the tactical RF-collection specialists operating in the modality the encryption headwind is compressing. Rayzone markets "full-scale, last-mile SIGINT capabilities" with "Tactical RF interception for 2G, 3G, 4G, and 5G networks" designed for "stationary, portable, and airborne deployment."⁹⁹ Septier markets a Guardian catcher "capturing up to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi" across "GSM, UMTS, LTE, and 5G," plus a DF/geolocation line.¹⁰⁰ Cellxion is the marketing-thin Wildcard, described via surveillance-industry-index listings as a UK cellular-interception supplier and held to descriptive treatment only.¹⁰¹

All three are private with no disclosed financials, so the Watch placement is structural and modality-driven, not a vendor-distress read. The RF-catcher modality's classic passive-content yield is eroding toward identity/DF/metadata and active/hybrid catchers as transport encryption hardens. And – as the same outside-verifiability limit that governs every RF specialist in this front – no vendor's "5G" listing is externally verifiable as live 5G-standalone content interception, the white space where, per market commentary, "no vendor has yet established a clearly dominant globally deployed commercial solution."⁸⁷ Watch signal: a named field-deployment reference, a demonstrated 5G-SA capability, or movement up-stack toward the forensic edge.

Also on Watch as a market-structure datapoint: **Magnet Forensics**, taken private by Thoma Bravo (~C\$1.8B headline; closed April 2023) and combined with Grayshift into an access-plus-analysis stack – a winner on capability but dark on disclosure since the take-private, so it cannot be graded on numbers.¹⁰²

Losers. No graded contender on this front earns a Losers label in this section. A vendor reaches this line only when a cited public event – a layoff specific to the tactical/forensic business, a missed quarter, a down-round, a named executive departure framed as distress, or a customer-churn disclosure – anchors a vendor-specific decline, per the casualty-evidence discipline. The public events on this front are momentum and consolidation, not failure. Cellebrite and MSAB are growing on disclosed financials; Magnet sits inside a private-equity portfolio with no public distress event; and the private RF specialists publish no figures that would support either a winner or a loser label. The structurally pressured force on this front is the never-graded remote-implant surveillance cohort (NSO, Candiru, Intellexa), which faces a tightening – and, as of December 2025, partly loosening – sanctions and export-control regime. That pressure is referenced as a market structure, named-outlet-attributed, and is deliberately *not* a Losers grade on any contender, because the cohort is carved out of grading and because no graded vendor's balance sheet is implicated.^{96 98} Quarterly refreshes will populate this line if a tactical-specific distress signal emerges on the public record.

3.7 The Campaign Ahead

Five watchlist items for H2 2026. Each names a public, observable threshold that would confirm or refute a claim above.

1. **The forensics-versus-catcher scorecard – Cellebrite/MSAB disclosure against the cellular-interception prints.** Signal to monitor: Cellebrite's next ARR/revenue disclosure and MSAB's next interim report, set against the next methodology-disclosed cellular-interception or IMSI-catcher market figure. Threshold for re-assessment: the forensics issuers sustaining low-to-mid-teens-or-better disclosed growth while the catcher-market CAGRs stay in the mid-single-to-high-single-digit band → Pattern Claim 1's edge-forensics ascendancy holds. Inverse threshold: Cellebrite ARR growth decelerating below ~12% across two consecutive quarters, or a publisher showing the cellular-interception CAGR re-accelerating above the forensics CAGR → the wire is reopening and the ascendancy read weakens. Primary source: Cellebrite SEC 6-K filings and IR releases, MSAB interim/year-end reports, Mordor / MarketsandMarkets methodology-disclosed updates.^{89 90 87 88}

Update (September 2026): the 2026-08-13 Q2 print held ARR growth at 21% but cut the FY2026 ARR growth outlook to 14–16% on “longer sales cycles”; the ~12% two-quarter inverse threshold is not tripped. Next print: Q3 2026, November.⁸⁹

2. **A named field-extraction reference win — the lab-to-field collapse made concrete.** Signal: a named government field-deployment reference for a kiosk, ruggedized, or drone-forensics product (MSAB XRY Frontline/Drone, Cellebrite ruggedized UFED), as distinct from homepage positioning. Threshold: one named field-deployment reference, or a further named field/edge form-factor launch from a forensics anchor, in H2 2026 → Pattern Claim 2’s lab-to-field collapse hardens into the front’s direction. Inverse: 2026 roadmaps re-centering on lab throughput with field form factors quietly retired, or a named legal/policy constraint cutting border device-search volumes → the collapse stays a niche of the lab market. Primary source: vendor newsrooms (MSAB, Cellebrite, Magnet), DHS/CBP/ICE procurement records, named-outlet field-forensics coverage.^{91 92 93 94} **Update (September 2026):** Cellebrite launched C-TEK, “a ruggedized, all-in-one mobile forensics field kit packaged in a standard-size backpack” for “units operating in disconnected environments” (2026-08-26) — a further named field/edge form-factor launch from a forensics anchor, meeting this signal’s threshold; no named government field-deployment reference was published with it.⁹³
3. **A further Entity-List action or a broader sanctions lift — which way the reckoning runs.** Signal: whether H2 2026 brings additional Entity-List additions or sanctions on surveillance-cohort entities, or instead extends the December 2025 partial Intellexa lift into broader relief and a looser export-control posture. Threshold: a new Entity-List/sanctions action, or visible export-licensing friction reshaping cross-border field-collection sale → Pattern Claim 3’s surveillance-market reckoning strengthens and budget keeps migrating to the accountable/forensic end. Inverse: a broader sanctions lift following the December 2025 partial action, with a loosening posture → the legitimacy pressure that has favored accountable vendors is relaxing. Primary source: BIS / Federal Register Entity-List postings, OFAC/Treasury designations, EU dual-use list updates, Access Now / named-outlet surveillance-accountability coverage.^{96 97 98}
4. **A named 5G-standalone-capable catcher launch or independent test.** Signal: a tactical RF vendor publishing a named government reference, or an independent test, asserting live identity collection against a 5G *standalone* network with SUCI encryption — not 5G non-standalone anchored to a 4G core. Threshold: one such named 5G-SA reference or independent test through H2 2026 → the “5G coverage is a checkbox over a 4G-anchored reality” read weakens and the catcher modality has a genuine path past the encryption headwind. Inverse: uniform “supports 5G” listings continuing with no named 5G-SA outcome → “5G” stays table-stakes positioning over an undisclosed standalone reality, and the catcher specialists stay on Watch. Primary source: vendor product pages (Rayzone, Septier), named-outlet RF/SIGINT coverage, market-research notes flagging the 5G-SA white space.^{99 100 87}
5. **An oversight ruling or warrant-law shift reshaping field-collection procurement.** Signal: a U.S. or EU/member-state legal or oversight development — a warrant-statute change for cell-site simulators, a border device-search ruling, or a domestic interception-authorization shift — that materially expands or constrains lawful field deployment. Threshold: a named ruling or statute that expands lawful field-extraction or catcher use → the demand surface under Pattern Claims 1 and 2 widens. Inverse: a named ruling that constrains border device searches or tightens catcher warrant

requirements → the most recurring, least-encryption-defeated demand surface in the front narrows. Primary source: U.S. Congress.gov / DOJ pen-register policy / court dockets, EU EPRS and member-state interception-law publications, named-outlet legal coverage.^{86 94}

References & Footnotes

1. Electronic Frontier Foundation, "Cell-Site Simulators / IMSI Catchers," Street Level Surveillance, <https://sls.eff.org/technologies/cell-site-simulators-imsi-catchers> (accessed 2026-06-14). Verbatim: "masquerade as legitimate cell-phone towers, tricking phones within a certain radius into connecting to the device rather than a tower"; "Passive devices, as a rule, do not transmit any signals. They work by plucking cellular transmissions out of the air"; active simulators "broadcast signals that are either stronger than the legitimate cell sites around them, or are made to appear stronger"; "If the cell-site simulator is able to downgrade the cellular connection to a 2G/GSM connection then it can potentially perform much more intrusive acts such as intercepting call metadata...the content of unencrypted phone calls and text messages."
2. Rayzone Group, "Proximity-Data Intelligence," <https://rayzone.com/solution/proximity-data-intelligence/> (accessed 2026-06-14). Verbatim: "Tactical RF interception for 2G, 3G, 4G, and 5G networks"; "Designed for stationary, portable, and airborne deployment." Company framing "last-mile SIGINT capabilities, empowering tactical teams with real-time situational awareness and on-site threat mitigation" and ArrowCell ("detects the presence of an IMSI catcher"; users can "disrupt and locate their attacker") per rayzone.com product/solution pages (accessed 2026-06-14).
3. Septier Communication Ltd., "Septier Guardian Airborne," <https://www.septier.com/portfolio-item/septier-guardian-airborne/> (accessed 2026-06-14).
4. Electronic Frontier Foundation, "Meet Rayhunter: A New Open Source Tool from EFF to Detect Cellular Spying," <https://www.eff.org/deeplinks/2025/03/meet-rayhunter-new-open-source-tool-eff-detect-cellular-spying> (dated March 2025, accessed 2026-06-14). Verbatim: "a new open source tool... to detect cellular spying."
5. MSAB, "Mobile Forensics in 2026: The Comprehensive Guide to Extraction, Analysis, and Trends," <https://www.msab.com/blog/mobile-forensics-comprehensive-guide/> (dated 28 April 2026, accessed 2026-06-14). Verbatim: "Mobile forensics is a branch of digital forensics relating to the recovery of digital evidence from a mobile device under forensically sound conditions"; logical ("the forensic tool communicates with the OS via API"); full file system ("elevate privileges to gain 'Root' access, allowing them to copy the entire file system and its decryption keys"); physical ("a bit-by-bit copy of the physical memory (NAND flash), this is the gold standard of extraction"); "If a device is in a BFU (Before First Unlock) state, the data is heavily encrypted"; "MSAB's latest innovation allows agencies to utilize high-power GPUs to accelerate the brute-force process. What used to take weeks can now be accomplished in hours"; "The trend in 2026 is moving extraction to the field."
6. Cellebrite, "Cellebrite UFED I Mobile Forensic Data Extraction Tool," <https://cellebrite.com/en/products/ufed/> (accessed 2026-06-14). Verbatim: "The Industry Standard for Lawfully Accessing and Collecting Digital Data"; "Don't let complicated locks, encryption barriers, deleted and unknown content prevent critical evidence from coming to light"; "full file system (FFS) and physical extractions"; "UFED Touch3 Ruggedized Tablet enables comprehensive data extraction collection capabilities anywhere, whether in the lab, a remote location, or in the field."
7. Septier Communication Ltd., Septier Guardian IMSI Catcher product descriptions, <https://www.septier.com/portfolio-item/septier-identities-catcher/> (accessed 2026-06-14; the product page has since migrated, and the described capability is confirmed against the dated Internet Archive Wayback capture, <https://web.archive.org/web/20260312080626/https://septier.com/portfolio-item/septier-identities-catcher/>, consistent with the re-confirmed Septier Guardian sourcing at⁵⁵). Described capability: "capturing up

- to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi"; extracts "cellular identities, location, and other information from GSM, UMTS, LTE, and 5G devices in its coverage area"; Septier Cellular Locator "a tactical location tracking solution that provides accurate location information for cellular devices, using advanced direction finders such as Septier Hunter."
8. Rayzone Group product descriptions (Piranha, Vegas), <https://rayzone.com/> and named-outlet coverage (Calcalist/CTech, calcalistech.com; Israel Defense, israeldefense.co.il) (accessed 2026-06-14). Piranha described as "an SDR (Software Defined Radio) collector of mobile phone identifiers (IMSI & IMEI) in 2G, 3G, 4G and 5G environments."
 9. CRFS, "Angle of Arrival / Direction Finding Techniques" and "Is your RF recording for SIGINT high-fidelity?," <https://pages.crfs.com/hubfs/whitepapers/Angle%20of%20Arrival-Direction%20Finding.pdf> and <https://www.crfs.com/blog/is-your-rf-recording-for-sigint-high-fidelity> (accessed 2026-06-14). TDOA "comparing the signal's arrival time at three or more nodes"; AoA "using two direction finding arrays."
 10. Wintego "WINT" / "CatchApp" named-outlet coverage: Andrea Fortuna, <https://andreafortuna.org/2016/10/04/wintego-launches-a-system-called-catchapp-that-is-able-to-spy-whatsapp-encrypted-communications/>; Techworm, <https://www.techworm.net/2016/09/catchapp-tool-can-siphon-encryption-whatsapp-messages-distance.html> (2016, accessed 2026-06-14). WINT described as "a tactical solution for extracting intelligence over WiFi networks... through Man-in-the-Middle (MitM) interception... small enough to fit into a backpack." No current dated self-positioning page surfaces; referenced for modality context only, not graded, and no operating-status claim is made.
 11. Magnet Forensics, Magnet AXIOM positioning, vendor and named-outlet coverage (Police1, <https://www.police1.com/>; BusinessWire, <https://www.businesswire.com/>) (accessed 2026-06-14). Positioning: examiners can "gather all digital evidence associated with a case... whether the evidence lived on a smartphone, the cloud, a computer, a USB drive, other removable media, or even IoT-related devices."
 12. Mordor Intelligence, "IMSI Catcher Market Size, Share & 2031 Growth Trends Report," <https://www.mordorintelligence.com/industry-reports/imsi-catcher-market> (accessed 2026-06-14). "5G SA-compatible interception, where no vendor has yet established a clearly dominant globally deployed commercial solution." (Single-firm estimate for an opaque B2G niche – directional, not authoritative.)
 13. IMSI-catcher market trend commentary (hybrid active/passive), Accio business trends, <https://www.accio.com/business/imsi-catcher-trends> (accessed 2026-06-14). "Hybrid IMSI catchers, which combine active and passive capabilities, are gaining traction and are expected to be the most lucrative design niche through 2030." (Third-party market commentary, directional.)
 14. "IMSI-catcher," Wikipedia, <https://en.wikipedia.org/wiki/IMSI-catcher> (accessed 2026-06-14), summarizing the 3GPP 5G design: the standard "attempts to fix IMSI-catching issues by encrypting the permanent identifier (now SUPI) and transmitting the SUCI," with residual "risks... through downgrade attacks and implementation flaws." (Tertiary source for a standards summary; the underlying SUPI/SUCI design is specified in 3GPP TS 33.501.)
 15. NIST Computer Security Resource Center, "Wassenaar Arrangement Control Implementation: Intrusion and Surveillance Items," https://csrc.nist.gov/CSRC/media/Presentations/Wassenaar-Arrangement-Control-Implementation-In/images-media/ispab_june-11_rrarog.pdf (accessed 2026-06-14), and Wassenaar Arrangement control-list summaries. "The Intrusion Software and IP Network Surveillance controls represent two distinct types of definitions under different sections of the Control List, with the former under Category 4 (Computers) and the latter under Category 5 Part 1 (Telecommunications)"; intrusion software "defined as software designed to covertly gain access to a computer or other networked device and, once inside, to extract or modify data or modify the execution path." December 2024 Plenary "adopted updated controls on intrusion software and IP-network surveillance systems."
 16. Mondaq / Akin Gump summaries of European Commission, "Guidelines on the Export of Cyber-surveillance Items Under Article 5 of Regulation 2021/821," <https://www.mondaq.com/unitedstates/export-controls-trade-investment-sanctions/1551428/> and <https://www.akingump.com/en/insights/alerts/guidelines-on-the-export-of-cyber-surveillance-items-under-article-5-of-regulation-2021-821> (EU guidelines issued 15 October 2024,

- accessed 2026-06-14). Article 5 “mandates exporters to seek authorization if they know or suspect that their cyber-surveillance items could be used for human rights abuses, even if these items aren’t explicitly covered by existing export controls.”
17. SIPRI, “Making the most of the EU catch-all control on cyber-surveillance exports,” <https://www.sipri.org/commentary/topical-background/2024/making-most-eu-catch-all-control-cyber-surveillance-exports> (2024, accessed 2026-06-14). Cyber-surveillance items defined as dual-use items “specially designed to enable the covert surveillance of natural persons by monitoring, extracting, collecting or analysing data from information and telecommunication systems.”
 18. US Bureau of Industry and Security interim final rule “Information Security Controls: Cybersecurity Items” (October 2021), Federal Register 2021-22774, <https://www.federalregister.gov/documents/2021/10/21/2021-22774/information-security-controls-cybersecurity-items>, summarized by trade counsel: Lexology, <https://www.lexology.com/library/detail.aspx?g=00cdbcce8-cafd-4f4a-ab99-9543bbfcf6cd>; Davis Wright Tremaine, <https://www.dwt.com/blogs/privacy--security-law-blog/2021/11/commerce-department-cybersecurity-export-controls> (accessed 2026-06-14). ECCN 5A001.j controls “IP network communications surveillance systems or equipment,” capturing “[a]nalysis at the application layer,” extraction of “selected metadata and application content,” and mapping “the relational network of an individual or of a group of people”; ECCN 4D004 controls “‘software’ ‘specially designed’ or modified for the generation, command and control, or delivery of ‘intrusion software.’”
 19. “Cell-Site Simulator Warrant Act of 2023” (H.R.6194, 118th Congress), <https://www.congress.gov/bill/118th-congress/house-bill/6194/text>; Project On Government Oversight issue brief, <https://www.pogo.org/fact-sheets/issue-brief-the-cell-site-simulator-warrant-act> (accessed 2026-06-14). Reintroduced November 2023 to “create clear warrant requirements for use of cell site simulators by any government agency.”
 20. NYPD, “Cell-Site Simulators: Impact and Use Policy” (24 November 2023), https://www.nyc.gov/assets/nypd/downloads/pdf/public_information/post-final/cell-site-simulators-nypd-Impact-and-use-policy_11.24.23.pdf (accessed 2026-06-14). Cell-site simulators “must be configured as pen registers, and may not be used to collect the contents of any communication, in accordance with 18 U.S.C. § 3127(3).”
 21. Research and Markets, “Lawful Interception Market - Global Forecast 2026-2032,” <https://www.researchandmarkets.com/reports/5674490/lawful-interception-market-global-forecast> (accessed 2026-06-14; re-checked 2026-09-11 — the page was re-issued September 2026 and now states: “The Lawful Interception Market is projected to reach USD 3.98 Billion in 2026. It is expected to continue growing at a CAGR of 9.47%, reaching USD 6.98 Billion by 2032”; the June figures are the prior edition). USD 6.24B (2024) → USD 7.50B (2025); ~20.12% CAGR (2025–2032) [prior edition]. URL re-pointed 2026-09-11 from report 5939301, whose own figures this report never printed.
 22. MarketsandMarkets, “Lawful Interception Market,” <https://www.marketsandmarkets.com/Market-Reports/lawful-interception-market-1264.html>; release <https://www.globenewswire.com/news-release/2026/05/25/3300654/0/en/lawful-interception-market-surges-to-24-60-billion-by-2032-cagr-20-1.html> (dated 2026-05-25, accessed 2026-06-14; vendor page re-checked 2026-09-11, Jun-2026 edition: “The global lawful interception market is expected to grow from USD 8.20 billion in 2026 to USD 20.58 billion by 2031, at a CAGR of 20.2%”, with law enforcement agencies “62.6% of the market in 2026”). USD 8.20B (2026) → USD 24.60B (2032); 20.1% CAGR (2026–2032) [2026-05-25 release, unchanged].
 23. Mordor Intelligence, “Lawful Interception Market,” <https://www.mordorintelligence.com/industry-reports/lawful-interception-market> (accessed 2026-06-14; figure re-verified live 2026-06-15). USD 5.47B (2025) → USD 6.47B (2026) → USD 14.98B (2031); 18.27% CAGR (2026–2031).
 24. MarketIntelto, “IMSI Catcher Market” (USD 400M (2024) → USD 950M (2033); 9.8% CAGR) — a single-firm directional estimate. This exact figure has circulated under more than one publisher/aggregator listing; Verified Market Research’s own current “IMSI Catcher Market” page

- (<https://www.verifiedmarketresearch.com/product/imsi-catcher-market/>) shows a different bracket (~USD 556.7M (2024) → ~USD 1,192.35M), so the \$400M→\$950M figure is carried as one directional single-firm estimate among several rather than as an authoritative publisher number (accessed 2026-06-14).
25. Mordor Intelligence, "Cellular Interception Market," <https://www.mordorintelligence.com/industry-reports/cellular-interception-market> (accessed 2026-06-14). USD 695.49M (2025) → USD 735.69M (2026) → USD 921.57M (2031); 4.61% CAGR (2026–2031). Same report: UAV-mounted interception payloads "extend tactical flexibility and range, particularly for border surveillance"; 198,100 outdoor US small cells by mid-2025; operators migrating to standalone 5G cores "must expose interception interfaces per ETSI TS 33.127 v18.11.0 standards"; 181 carriers across 73 nations invested by late 2025; "5G NR is projected to record the fastest 5.36% CAGR through 2031" on this cellular-interception base (re-checked 2026-09-11 — the ~11.44% 5G-NR figure sits on Mordor's separate IMSI-catcher report, not this one, and was re-pointed accordingly on 2026-09-11); US Army USD 95.4M tactical communications modernization; US DHS USD 43.8B surveillance allocation FY2026.
 26. Mordor Intelligence, "IMSI Catcher Market Size, Share & 2031 Growth Trends Report," <https://www.mordorintelligence.com/industry-reports/imsi-catcher-market> (accessed 2026-06-14). USD 182.57M (2025) → USD 207.68M (2026) → USD 306.47M (2031); 8.08% CAGR. Same report (re-checked 2026-09-11): "5G NR solutions are expected to advance at a 11.44% CAGR through 2031."
 27. MarketsandMarkets, "Mobile Forensics Market," <https://www.marketsandmarkets.com/Market-Reports/mobile-forensics-market-29093950.html> (accessed 2026-06-14). USD 5.72B (2026) → USD 9.99B (2031); 11.8% CAGR (2026–2031).
 28. Precedence Research, "Digital Forensics Market," <https://www.precedenceresearch.com/digital-forensics-market> (accessed 2026-06-14). USD 13.46B (2025) → USD 15.50B (2026) → USD 53.65B (2035); 14.83% CAGR (2026–2035); mobile forensics stated to dominate revenue share in 2025.
 29. Mordor Intelligence, "Digital Forensics Market," <https://www.mordorintelligence.com/industry-reports/digital-forensics-market> (accessed 2026-06-14; page re-issued 2026-07-23 and re-checked 2026-09-11: "The digital forensics market size is projected to expand from USD 7.51 billion in 2025 and USD 8.46 billion in 2026 to USD 15.37 billion by 2031, registering a 12.68% CAGR between 2026 and 2031"; the June figures are the prior edition). USD 14.85B (2025) → USD 31.74B (2030); 16.5% CAGR (2025–2030) [prior edition]. "the top five vendors, OpenText, Cellebrite, Magnet Forensics, Exterro, and MSAB, accounted for about 38% of 2025 revenue." (Single-firm estimate; directional.)
 30. MSAB, "MSAB publishes annual report for 2025," <https://www.msab.com/updates/msab-publishes-annual-report-for-2025/> and "Year-end report 2025," <https://www.msab.com/updates/year-end-report-2025> (accessed 2026-06-14). Full-year 2025 net sales SEK 461.8M, up 14.1% (currency-adjusted 19.7%); operating profit (EBIT) SEK 67.8M (14.7% margin); profit after tax SEK 49.2M; described as "the most successful year in the company's history." Top-five digital-forensics vendor concentration ~38% of 2025 revenue per the same annual report.
 31. Fortune Business Insights, "Signals Intelligence Market," <https://www.fortunebusinessinsights.com/signals-intelligence-market-115665> (accessed 2026-06-14). USD ~19.84B (2025) → USD 29.77B (2034); 4.61% CAGR.
 32. Future Market Insights, "Signals Intelligence (SIGINT) Market," <https://www.futuremarketinsights.com/reports/signals-intelligence-sigint-market> (accessed 2026-06-14). USD ~17.7B (2025) → USD 30.0B (2035); 5.4% CAGR.
 33. Cellebrite, "Cellebrite Announces First-Quarter 2026 Results," dated 2026-05-14, <https://www.prnewswire.com/news-releases/cellebrite-announces-first-quarter-2026-results-302772161.html>; Form 6-K <https://www.sec.gov/Archives/edgar/data/0001854587/000121390026056220/ea029072401ex99-1.htm> (accessed 2026-06-14). Q1 FY2026 (three months ended 2026-03-31): total revenue USD 128.3M (+19% YoY); subscription revenue USD 117.9M (+23%); ARR USD 493.0M (+21% YoY); dollar-based net revenue retention 115%; GAAP gross margin 82.5%. Full-year 2026 guidance as issued 2026-05-14: ARR USD 567–573M (18–19% growth); revenue USD 565–571M (19–20% growth). Q2 FY2026 (three months ended 2026-06-30, reported 2026-08-13): total revenue USD 131.1M (+16% YoY); subscription revenue USD 119.5M (+16%); ARR USD 507.8M (+21% YoY); dollar-based net revenue retention 117%; GAAP gross margin 80.8%; GAAP net income USD 6.4M; adjusted

- EBITDA USD 31.8M (24.2% margin). Full-year 2026 outlook lowered on 2026-08-13 to ARR USD 550–560M (14–16% growth) and revenue USD 555–561M (17–18% growth), with the adjusted-EBITDA target raised to USD 153–159M – Cellebrite Q2 2026 results, Exhibit 99.1 to Form 6-K furnished 2026-08-13, <https://www.sec.gov/Archives/edgar/data/1854587/000121390026088681/ea030188001ex99-1.htm> (accessed 2026-09-11). NASDAQ: CLBT – trading-position disclaimer applies; re-verify against next 6-K at authoring.
34. Thoma Bravo and Magnet Forensics, "Magnet Forensics Announces Closing of Plan of Arrangement with Thoma Bravo," closing announcement dated 2023-04-06, <https://www.thomabravo.com/press-releases/magnet-forensics-announces-closing-of-plan-of-arrangement-with-thoma-bravo> (accessed 2026-08-22). The release confirms the Magnet arrangement closed and describes bringing Magnet Forensics and Grayshift together; it says Grayshift partnered with Thoma Bravo in 2022 but does not state a Grayshift ownership percentage. McMillan LLP, transaction counsel, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/>, supports the approximately C\$1.8B Magnet transaction value. TechCrunch, 2023-01-20, <https://techcrunch.com/2023/01/20/thoma-bravo-agrees-to-acquire-digital-forensics-firm-magnet-forensics-for-over-1b>, and GovTech, <https://www.govtech.com/biz/magnet-forensics-to-merge-with-grayshift-after-1-35b-buyout>, provide named-outlet USD-value corroboration (all accessed 2026-08-22). The C\$1.8B / USD-value reporting belongs to Magnet's take-private, not to a purchase of Grayshift. Thoma Bravo, "Grayshift Announces Strategic Investment from Thoma Bravo," announcement dated 2022-06-21, <https://www.thomabravo.com/press-releases/grayshift-announces-strategic-investment-from-thoma-bravo> (accessed 2026-08-22). The release characterizes the transaction as a strategic growth investment, says financial terms were not disclosed, says existing investors would remain shareholders, and says the investment was subject to regulatory approvals. It does not establish majority control or a July 2022 close.
35. Oxygen Forensics, "3 Solutions for Mobile Forensics Challenges in 2025," <https://www.oxygenforensics.com/blog/mobile-forensics-challenges-2025-solutions/> (accessed 2026-06-14). "the widespread use of end-to-end encryption by apps like WhatsApp, Signal, and Telegram makes data extraction increasingly challenging."
36. European Parliamentary Research Service, "Access to data for law enforcement and digital forensics," EPRS PE 775879 (2025), [https://www.europarl.europa.eu/RegData/etudes/BRIE/2025/775879/EPRS_BRI\(2025\)775879_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2025/775879/EPRS_BRI(2025)775879_EN.pdf) (accessed 2026-06-14). Frames device-level forensic access as the law-enforcement response when communications interception is defeated by encryption.
37. FedScoop, "DHS units to re-up contract with controversial mobile device data extractor," <https://fedscoop.com/dhs-cellebrite-privacy-drones-data-mobile-devices-ice/> (accessed 2026-06-14); AFSC Investigate, <https://investigate.afsc.org/company/cellebrite> (accessed 2026-06-14). CBP searched 47,000 electronic devices in 2024, up from ~8,500 in 2015; DHS/ICE+HSI planned a five-year IDIQ with a USD 100M ceiling for Cellebrite; between 2008 and April 2025 ICE awarded Cellebrite at least 213 contracts worth over USD 48.6M; CBP had eight Cellebrite contracts set to end between July 2025 and April 2026.
38. FedScoop / AFSC Investigate (as in ³⁷), accessed 2026-06-14.
39. Akin Gump / Lexology on European Commission, "Guidelines on the Export of Cyber-surveillance Items Under Article 5 of Regulation 2021/821," <https://www.akingump.com/en/insights/alerts/guidelines-on-the-export-of-cyber-surveillance-items-under-article-5-of-regulation-2021-821> (EU guidelines issued 15 October 2024, accessed 2026-06-14). Controlled cyber-surveillance items include "equipment for covertly extracting communication content and metadata that is transmitted over the air via wireless communication, and radiofrequency monitoring equipment." US implemented EAR controls on cybersecurity items in March 2022.
40. European Commission, "Exporting dual-use items," https://policy.trade.ec.europa.eu/help-exporters-and-importers/exporting-dual-use-items_en; "2025 update of EU control list of dual-use items," https://policy.trade.ec.europa.eu/news/2025-update-eu-control-list-dual-use-items-2025-09-08_en (dated

- 2025-09-08, accessed 2026-06-14). Annex I implements regime controls into the EU's binding dual-use list; Article 5 adds a catch-all for cyber-surveillance items not on the list where there is awareness of potential misuse.
41. OrangeSlices AI, "Contract Award \$6M DOJ/FBI Cellebrite Unlock and Extraction Software," <https://orangeslices.ai/contract-award-6m-doj-fbi-cellebrite-unlock-and-extraction-software/> (accessed 2026-06-14). FBI/DOJ awarded Cellebrite a contract valued at USD 5,717,319 for unlock and extraction software, start 2025-09-30, completion 2030-09-29, via the NASA SEWP vehicle.
 42. MSAB, "MSAB Receives Order from United States Government Agency," <https://www.msab.com/updates/msab-receives-order-from-united-states-government-agency> (accessed 2026-06-14); Cellebrite federal-vehicle availability per Officer.com press release, <https://www.officer.com/command-hq/technology/press-release/12097706/> (accessed 2026-06-14).
 43. Cellebrite, "Cellebrite UFED I Mobile Forensic Data Extraction Tool," <https://cellebrite.com/en/ufed/> (and <https://cellebrite.com/en/products/ufed/>), accessed/re-fetched 2026-06-14. Verbatim: "The Industry Standard for Lawfully Accessing and Collecting Digital Data"; "Access critical digital evidence lawfully from feature and smartphones, drones, SIM cards, SD cards, GPS devices, and more"; "Don't let complicated locks, encryption barriers, deleted and unknown content prevent critical evidence from coming to light"; "Access to containerized and encrypted data to uncover every device's Full File System (FFS) and protected data"; "Extract data from anywhere, whether in the lab, a remote location, or in the field"; "Forensically Sound Collection."
 44. Cellebrite, "Cellebrite Spring 2026 Release I Digital Forensics Updates," <https://cellebrite.com/en/products/launches-releases/spring-release-2026/>, accessed 2026-06-14. Verbatim: "expands device access capabilities across the widest range of iOS and Android devices and operating systems, including support for iPhone 17 and iOS 26"; "strengthens AI-powered decoding." Cellebrite Genesis (AI investigation surface): public-safety launch June 2026; enterprise general availability and wider regional availability announced 2026-08-19 – verbatim: "Cellebrite Genesis is expanding to more global markets and is now generally available for enterprise organizations" – Cellebrite release, <https://cellebrite.com/en/resources/press-releases/cellebrite-genesis-for-enterprise-now-generally-available-regional-availability-expands-to-more-global-markets/> (accessed 2026-09-11).
 45. Cellebrite DI Ltd. (NASDAQ: CLBT), Q2 2026 results (quarter ended 2026-06-30, reported 2026-08-13) – revenue USD 131.1M (+16% YoY), subscription revenue USD 119.5M (+16%), ARR USD 507.8M (+21%), adjusted EBITDA USD 31.8M (24.2% margin), GAAP net income USD 6.4M, GAAP gross margin 80.8%, net revenue retention 117%; full-year 2026 outlook lowered the same day to ARR USD 550–560M (14–16% growth) and revenue USD 555–561M, with the adjusted-EBITDA target raised to USD 153–159M – Exhibit 99.1 to Form 6-K furnished 2026-08-13, <https://www.sec.gov/Archives/edgar/data/1854587/000121390026088681/ea030188001ex99-1.htm> (accessed 2026-09-11). CEO succession announced the same day in a separate Form 6-K, in the issuer's own words: "Shiven Ramji was appointed Chief Executive Officer as part of a planned succession" and "effective as of August 13, 2026, Mr. Hogan resigned from his position on the board of directors," https://www.sec.gov/Archives/edgar/data/1854587/000121390026089232/ea0302091-6k_cellebrite.htm (accessed 2026-09-11). Prior dated quarter – Q1 2026 results: revenue USD 128.3M (+19% YoY), subscription revenue USD 117.9M (+23%), ARR USD 493M (+21%), adjusted EBITDA USD 30.6M (+29%), GAAP net income USD 10.9M; Q2 2026 guidance as issued 2026-05-14 ARR USD 510M–513M, revenue USD 130M–133M; drone-forensics portfolio expansion plus SCG Canada acquisition. Cellebrite Q1 2026 press release, furnished as Exhibit 99.1 to Cellebrite DI Ltd.'s Form 6-K (CIK 0001854587; filed 2026-05-14), <https://www.sec.gov/Archives/edgar/data/1854587/000121390026056220/ea029072401ex99-1.htm>; SEC filing detail, <https://www.sec.gov/Archives/edgar/data/1854587/0001213900-26-056220-index.htm>; named-outlet coverage: TipRanks, "Cellebrite Posts Strong Q1 2026 Growth, Expands AI and Drone Forensics Portfolio," <https://www.tipranks.com/news/company-announcements/cellebrite-posts-strong-q1-2026-growth-expands-ai-and-drone-forensics-portfolio>; GuruFocus, "Cellebrite DI Ltd (CLBT) Q1 2026 Earnings Call Highlights," <https://www.gurufocus.com/news/8860537/> (accessed 2026-08-22). CLBT is a public issuer – the front-matter trading-position disclosure applies.

46. MSAB, corporate homepage, "MSAB — Trusted Partner in Digital Forensics | XAMN & XRY," <https://www.msab.com/>, accessed 2026-06-14. Verbatim: "Trusted Partner in Digital Forensics."
47. MSAB, "Mobile Forensics in 2026: The Comprehensive Guide to Extraction, Analysis, and Trends," <https://www.msab.com/blog/mobile-forensics-comprehensive-guide/>, dated 28 April 2026, accessed 2026-06-14. Verbatim: "The trend in 2026 is moving extraction to the field"; "XRY Frontline (Kiosk, Tablet, Express)" deployed "at police stations and borders"; the extraction ladder (logical "communicates with the OS via API" / full file system "elevate privileges to gain 'Root' access" / physical "a bit-by-bit copy of the physical memory (NAND flash), this is the gold standard"); "If a device is in a BFU (Before First Unlock) state, the data is heavily encrypted"; "MSAB's latest innovation allows agencies to utilize high-power GPUs to accelerate the brute-force process. What used to take weeks can now be accomplished in hours."
48. MSAB (Micro Systemation AB, Nasdaq Stockholm), "Year-end report 2025," <https://www.msab.com/updates/year-end-report-2025> and <https://newsroom.msab.com/about-us/our-company/newsroom/press-releases/2026/year-end-report-2025/>, accessed 2026-06-14. Full-year 2025 net sales SEK 462M (+14.1%; +19.7% currency-adjusted); operating profit (EBIT) SEK 68M (14.7% margin); Q4 net sales SEK 140.6M (+19.6%).
49. Magnet Forensics, "Magnet Axiom | Digital Forensic Software," <https://www.magnetforensics.com/products/magnet-axiom/>, accessed/re-fetched 2026-06-14. Verbatim: "Bring together evidence recovery, analysis, and insight"; "Recover, analyze and report on data from mobile, computer, cloud, and vehicle sources"; "Access the most data possible, see the full story, and reduce time to evidence"; "Bring computer, mobile, cloud, vehicle, and acquired data into a single Axiom case"; "Access industry-leading artifact coverage."
50. Magnet Forensics, "Magnet User Summit 2026" / "Magnet Virtual Summit 2026," <https://www.magnetforensics.com/blog/magnet-user-summit-2026-heres-your-presentation-catalog/>; <https://www.magnetforensics.com/magnet-virtual-summit-2026-replays/> (accessed 2026-06-14) — evidence of the practitioner community.
51. Thoma Bravo and Magnet Forensics, "Magnet Forensics Announces Closing of Plan of Arrangement with Thoma Bravo," closing announcement dated 2023-04-06, <https://www.thomabravo.com/press-releases/magnet-forensics-announces-closing-of-plan-of-arrangement-with-thoma-bravo> (accessed 2026-08-22). The release confirms the Magnet arrangement closed and describes bringing Magnet Forensics and Grayshift together; it says Grayshift partnered with Thoma Bravo in 2022 but does not state a Grayshift ownership percentage. McMillan LLP, transaction counsel, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/>, supports the approximately C\$1.8B Magnet transaction value. TechCrunch, 2023-01-20, <https://techcrunch.com/2023/01/20/thoma-bravo-agrees-to-acquire-digital-forensics-firm-magnet-forensics-for-over-1b>, and GovTech, <https://www.govtech.com/biz/magnet-forensics-to-merge-with-grayshift-after-1-35b-buyout>, provide named-outlet USD-value corroboration (all accessed 2026-08-22). The C\$1.8B / USD-value reporting belongs to Magnet's take-private, not to a purchase of Grayshift. Thoma Bravo, "Grayshift Announces Strategic Investment from Thoma Bravo," announcement dated 2022-06-21, <https://www.thomabravo.com/press-releases/grayshift-announces-strategic-investment-from-thoma-bravo> (accessed 2026-08-22). The release characterizes the transaction as a strategic growth investment, says financial terms were not disclosed, says existing investors would remain shareholders, and says the investment was subject to regulatory approvals. It does not establish majority control or a July 2022 close.
52. Rayzone Group, "Proximity-Data Intelligence," <https://rayzone.com/solution/proximity-data-intelligence/>, re-fetched 2026-06-14. Verbatim: "full-scale, last-mile SIGINT capabilities"; "Tactical RF interception for 2G, 3G, 4G, and 5G networks"; "Designed for stationary, portable, and airborne deployment"; "Supporting fixed-site defense, mobile missions, and wide-area coverage"; "Tactical awareness. Real-time control."; "Real-Time Situational Awareness at the Tactical Edge"; "Operates discreetly to avoid detection."

53. Rayzone Group product descriptions (Piranha SDR IMSI-catcher 2G/3G/4G/5G; ArrowCell counter-IMSI-catcher), <https://rayzone.com/> and named-outlet coverage (CTech / Calcalist, <https://www.calcalistech.com/ctech/articles/0,7340,L-3884553,00.html>), accessed 2026-06-14. Piranha described as “an SDR (Software Defined Radio) collector of mobile phone identifiers (IMSI & IMEI)” for 2G–5G environments; ArrowCell for “Active GSM Elimination” / IMSI-catcher detection.
54. Rayzone Group company profile / client-geography color (Asia, Central America, Europe), treated as named-outlet/agggregator-sourced and attributed, not asserted: CTech, “The Rayzone Group’s secret cyber intelligence activities revealed,” <https://www.calcalistech.com/ctech/articles/0,7340,L-3884553,00.html>; Tracxn company profile, <https://tracxn.com/d/companies/rayzone/>; Crunchbase, <https://www.crunchbase.com/organization/rayzone-group> (accessed 2026-06-14). Funding/ownership carried as undisclosed.
55. Septier Communication Ltd., Septier Guardian IMSI Catcher + Cellular Locator product pages, <https://www.septier.com/portfolio-item/septier-identities-catcher/> and <https://www.septier.com/Septier%20Guardian%20IMSI%20Catcher>, re-confirmed 2026-06-14. Verbatim: “capturing up to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi”; extracts “cellular identities, location, and other information from GSM, UMTS, LTE, and 5G devices in its coverage area”; “available in various configurations to meet a range of tactical requirements”; Cellular Locator “a tactical location tracking solution that provides accurate location information for cellular devices, using advanced direction finders such as Septier Hunter.”
56. Septier Communication Ltd., “Septier Guardian Airborne,” <https://www.septier.com/portfolio-item/septier-guardian-airborne/>; “Intelligence Agencies” / “Police & Counter Terrorism” audience pages, <https://www.septier.com/portfolio-item/636/> and <https://www.septier.com/portfolio-item/police-counter-terrorism/> (accessed 2026-06-14). UAV-mounted Guardian variant “can extract 11 unique cellular identifiers for each detected device including IMSI, IMEI and more”; named buyer types: intelligence agencies, police, counter-terrorism units.
57. Cellxion (UK) cellular-interception / IMSI-catcher equipment (GX/UGX range), via the Surveillance Industry Index / Privacy International transparency listings and named directory descriptions (accessed 2026-06-14). *Named-outlet/index description, explicitly attributed as third-party characterization rather than vendor self-positioning; Wildcard caution applies — descriptive only, no pattern claim, no negative claim.*
58. Cognyte, “Border Protection / Maritime Intelligence” page, <https://www.cognyte.com/border-protection-maritime-intelligence-na/> (directly fetched, verbatim; accessed 2026-06-16). Verbatim: “Cognyte’s Border Protection solution takes a multi-layered approach by analyzing existing border sensors (cameras, radar, satellites and more) and fusing together with SIGINT, OSINT and VISINT sources.” Cognyte-era. This is this front’s *distinct primary pillar* (cross-front ledger); it is not reused as a primary pillar in any other front.
59. Tier-1 military-intelligence agency contract, ~USD 5M, described as deploying “a tactical SIGINT solution for operational missions” — BusinessWire / StockTitan, dated 2025-11-19 (named-outlet release, accessed 2026-06-16). Cognyte-era. Carried as corroboration on this front for the tactical-SIGINT positioning; the deal value and counterparty descriptor are quoted from the dated release, not asserted on this report’s authority.
60. VANTAGE trademark, USPTO Serial 78417244 / Reg. No. 3248122 (filed 2004-05-12; registered 2007-05-29) — goods (full registered clause): “Telecommunication surveillance software for the integrated interception, collection, monitoring, analyzing, searching, tracking and reporting of data and voice communications.” The USPTO Assignment Center abstract records an assignment of the entire interest in VANTAGE (Serial 78417244 / Reg. 3248122) from Verint Systems Inc. to Cognyte Software Ltd., executed 1 February 2021 and recorded 21 October 2021 at reel/frame 7484/0094. <https://assignmentcenter.uspto.gov/search/trademark/abstract%3FserialNumber%3D78417244> (accessed 2026-09-13). This dated record is cited as lineage, not as verification of current ownership. Carried as National-Agencies-weighted *corroboration only* in this front, never this front’s primary pillar; the goods-description wording is **confirmed** verbatim against the USPTO record; serial and registration number confirmed.

61. Mordor Intelligence, "IMSI Catcher Market Size, Share & 2031 Growth Trends Report," <https://www.mordorintelligence.com/industry-reports/imsi-catcher-market> (accessed 2026-06-14). IMSI-catcher segment "USD 207.68 million in 2026 ... CAGR of 8.08% to reach USD 306.47 million by 2031"; named majors "Rohde and Schwarz GmbH and Co KG, Septier Communication Ltd., L3Harris Technologies, Inc., Rayzone Group Ltd. and PKI Electronic Intelligence GmbH"; "5G SA-compatible interception, where no vendor has yet established a clearly dominant globally deployed commercial solution." (Single-firm estimate for an opaque B2G niche – directional, not authoritative; sizing owned by the Terrain section.)
62. Wintego "WINT" / "CatchApp" Wi-Fi man-in-the-middle system, named-outlet historical coverage (Andrea Fortuna, <https://andreafortuna.org/2016/10/04/wintego-launches-a-system-called-catchapp-that-is-able-to-spy-whatsapp-encrypted-communications/>; Techworm, <https://www.techworm.net/2016/09/catchapp-tool-can-siphon-encryption-whatsapp-messages-distance.html>, 2016) (accessed 2026-06-14). WINT described as "a tactical solution for extracting intelligence over WiFi networks ... through Man-in-the-Middle (MitM) interception ... small enough to fit into a backpack." No current dated self-positioning page surfaces; Wintego is referenced for Wi-Fi-MitM modality context only, not graded, and no operating-status claim is made.
63. 2026 forensics product-launch commentary (Cellebrite Premium Unlock Jan 2026; MSAB XRY Drone Nov 2025), via third-party aggregator/comparison pages (Gitnux, "Best Cell Phone Forensics Software 2026," <https://gitnux.org/best/cell-phone-forensics-software/>; Sherlock Forensics, <https://www.sherlockforensics.com/blog/cellebrite-vs-magnet-axiom-2026.html>) (accessed 2026-06-14). Aggregator commentary; the individual product-launch dates are carried as aggregator-attributed, not asserted on this report's authority.
64. MSAB, "Mobile Forensics in 2026: The Comprehensive Guide to Extraction, Analysis, and Trends," <https://www.msab.com/blog/mobile-forensics-comprehensive-guide/> (dated 28 April 2026, accessed 2026-06-14). Verbatim: "The trend in 2026 is moving extraction to the field"; "XRY Frontline (Kiosk, Tablet, Express)" deployed "at police stations and borders"; "MSAB's latest innovation allows agencies to utilize high-power GPUs to accelerate the brute-force process. What used to take weeks can now be accomplished in hours."
65. Cellebrite, "Cellebrite UFED I Mobile Forensic Data Extraction Tool," <https://cellebrite.com/en/ufed/> (accessed/re-fetched 2026-06-14). Verbatim: "Extract data from anywhere, whether in the lab, a remote location, or in the field."
66. Cellebrite DI Ltd. (NASDAQ: CLBT), Q1 2026 results – drone-forensics portfolio expansion noted in Q1 FY2026 commentary (reported 2026-05-14). Cellebrite Q1 2026 press release (SEC 6-K / 8-K exhibit) and named-outlet coverage: TipRanks, "Cellebrite Posts Strong Q1 2026 Growth, Expands AI and Drone Forensics Portfolio," <https://www.tipranks.com/news/company-announcements/cellebrite-posts-strong-q1-2026-growth-expands-ai-and-drone-forensics-portfolio> (accessed 2026-06-14). CLBT is a public issuer – trading-position disclaimer applies.
67. Thoma Bravo and Magnet Forensics, "Magnet Forensics Announces Closing of Plan of Arrangement with Thoma Bravo," closing announcement dated 2023-04-06, <https://www.thomabravo.com/press-releases/magnet-forensics-announces-closing-of-plan-of-arrangement-with-thoma-bravo> (accessed 2026-08-22). The release confirms the Magnet arrangement closed and describes bringing Magnet Forensics and Grayshift together; it says Grayshift partnered with Thoma Bravo in 2022 but does not state a Grayshift ownership percentage. McMillan LLP, transaction counsel, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/>, supports the approximately C\$1.8B Magnet transaction value. TechCrunch, 2023-01-20, <https://techcrunch.com/2023/01/20/thoma-bravo-agrees-to-acquire-digital-forensics-firm-magnet-forensics-for-over-1b>, and GovTech, <https://www.govtech.com/biz/magnet-forensics-to-merge-with-grayshift-after-1-35b-buyout>, provide named-outlet USD-value corroboration (all accessed 2026-08-22). The C\$1.8B / USD-value reporting belongs to Magnet's take-private, not to a purchase of Grayshift. Thoma Bravo, "Grayshift Announces Strategic Investment from Thoma Bravo," announcement dated 2022-06-21, <https://www.thomabravo.com/press-releases/grayshift-announces-strategic-investment-from-thoma-bravo>

- (accessed 2026-08-22). The release characterizes the transaction as a strategic growth investment, says financial terms were not disclosed, says existing investors would remain shareholders, and says the investment was subject to regulatory approvals. It does not establish majority control or a July 2022 close.
68. McMillan LLP, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/> (transaction value reported as C\$1.8 billion); Financier Worldwide, "Magnet Forensics sold to Thoma Bravo for \$1.3bn," <https://www.financierworldwide.com/magnet-forensics-sold-to-thoma-bravo-for-13bn> (US\$1.3bn per-share equity framing; CA\$44.25/share) (accessed 2026-06-14).
69. Mordor Intelligence, "IMSI Catcher Market Size, Share & 2031 Growth Trends Report," <https://www.mordorintelligence.com/industry-reports/imsi-catcher-market> (accessed 2026-06-14). Verbatim: "5G SA-compatible interception, where no vendor has yet established a clearly dominant globally deployed commercial solution." (Single-firm estimate for an opaque B2G niche – directional, not authoritative; sizing owned by the Terrain section.)
70. US Department of Commerce, Bureau of Industry and Security – Entity List additions: NSO Group and Candiru (4 November 2021, "presumption of denial"; determination that they "developed and supplied spyware to foreign governments" used to "maliciously target government officials, journalists, businesspeople, activists, academics, and embassy workers"), Federal Register, "Addition of Certain Entities to the Entity List," <https://www.federalregister.gov/documents/2021/11/04/2021-24123/addition-of-certain-entities-to-the-entity-list>; BIS press release, <https://www.bis.gov/press-release/commerce-adds-nso-group-other-foreign-companies-entity-list-malicious-cyber-activities>; Intellexa/Cytrox added 18 July 2023, CyberScoop, "Commerce Department blacklists spyware companies Cytrox and Intellexa," <https://cyberscoop.com/commerce-department-blacklists-spyware-companies/> (accessed 2026-06-14).
71. EU Dual-Use Regulation 2021/821 Article 5 cyber-surveillance catch-all; European Commission "Guidelines on the Export of Cyber-surveillance Items Under Article 5 of Regulation 2021/821" issued 15 October 2024. Summaries: Akin Gump, <https://www.akingump.com/en/insights/alerts/guidelines-on-the-export-of-cyber-surveillance-items-under-article-5-of-regulation-2021-821>; SIPRI, <https://www.sipri.org/commentary/topical-background/2024/making-most-eu-catch-all-control-cyber-surveillance-exports> (accessed 2026-06-14).
72. Cellebrite, "Cellebrite Reports Q1 2026 Results: ARR Reaches \$493 Million, Up 21% Year-Over-Year," reported 2026-05-14 (qtr ended 2026-03-31), via citybiz / Cellebrite IR, <https://www.citybiz.co/article/846282/cellebrite-reports-q1-2026-results-arr-reaches-493-million-up-21-year-over-year/> (ARR \$493M +21%; total revenue \$128.3M +19%; subscription revenue \$117.9M +23%; adj. EBITDA \$30.6M +29%; net revenue retention 115%; Q2 ARR guide \$510M–\$513M; FedRAMP High; federal pipeline +35%). Superseded as the current dated print by the Q2 2026 results (quarter ended 2026-06-30, reported 2026-08-13): ARR \$507.8M, +21% YoY; total revenue \$131.1M, +16%; subscription revenue \$119.5M, +16%; adj. EBITDA \$31.8M (24.2% margin); net revenue retention 117%; GAAP net income \$6.4M – Exhibit 99.1 to Form 6-K furnished 2026-08-13, <https://www.sec.gov/Archives/edgar/data/1854587/000121390026088681/ea030188001ex99-1.htm> (accessed 2026-09-11); the Q1 figures above stay as the prior dated quarter. CLBT is a public issuer – trading-position disclaimer applies.
73. Cellebrite DI Ltd., Form 6-K (Q1 FY2026 earnings press release exhibit), SEC, <https://www.sec.gov/Archives/edgar/data/0001854587/000121390026056220/ea029072401ex99-1.htm> (filing corroboration of Q1 FY2026 figures).
74. MSAB (Micro Systemation AB), "Year-end report 2025," for the year ended 2025-12-31, <https://www.msab.com/updates/year-end-report-2025/> (net sales SEK 462M, +14.1% YoY; currency-adjusted +19.7%; operating profit/EBIT SEK 68M, 14.7% margin; profit after tax SEK 49.2M; EPS SEK 2.66; proposed dividend SEK 1.60/share; "the most successful year in the company's history").
75. Thoma Bravo, "Magnet Forensics Announces Closing of Plan of Arrangement with Thoma Bravo," closing 2023-04-06, <https://www.thomabravo.com/press-releases/magnet-forensics-announces-closing-of-plan-of-arrangement-with-thoma-bravo> (CA\$44.25 per subordinate voting share; aggregate cash consideration

- ~CA\$1,044,155,363; rollover shares ~CA\$627,880,695; Grayshift Co-Founder/CEO David Miles on combining Magnet Forensics and Grayshift).
76. McMillan LLP, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/> (transaction value reported as C\$1.8 billion); Financier Worldwide, "Magnet Forensics sold to Thoma Bravo for \$1.3bn," <https://www.financierworldwide.com/magnet-forensics-sold-to-thoma-bravo-for-13bn> (US\$1.3bn per-share equity framing; CA\$44.25/share, ~15% premium to 19-Jan-2023 close) (accessed 2026-06-14).
 77. Thoma Bravo and Magnet Forensics, "Magnet Forensics Announces Closing of Plan of Arrangement with Thoma Bravo," closing announcement dated 2023-04-06, <https://www.thomabravo.com/press-releases/magnet-forensics-announces-closing-of-plan-of-arrangement-with-thoma-bravo> (accessed 2026-08-22). The release confirms the Magnet arrangement closed and describes bringing Magnet Forensics and Grayshift together; it says Grayshift partnered with Thoma Bravo in 2022 but does not state a Grayshift ownership percentage. McMillan LLP, transaction counsel, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/>, supports the approximately C\$1.8B Magnet transaction value. TechCrunch, 2023-01-20, <https://techcrunch.com/2023/01/20/thoma-bravo-agrees-to-acquire-digital-forensics-firm-magnet-forensics-for-over-1b>, and GovTech, <https://www.govtech.com/biz/magnet-forensics-to-merge-with-grayshift-after-1-35b-buyout>, provide named-outlet USD-value corroboration (all accessed 2026-08-22). The C\$1.8B / USD-value reporting belongs to Magnet's take-private, not to a purchase of Grayshift. Thoma Bravo, "Grayshift Announces Strategic Investment from Thoma Bravo," announcement dated 2022-06-21, <https://www.thomabravo.com/press-releases/grayshift-announces-strategic-investment-from-thoma-bravo> (accessed 2026-08-22). The release characterizes the transaction as a strategic growth investment, says financial terms were not disclosed, says existing investors would remain shareholders, and says the investment was subject to regulatory approvals. It does not establish majority control or a July 2022 close.
 78. Crunchbase, Rayzone Group company profile (no disclosed priced funding round; revenue bands are aggregator-directional, not load-bearing), <https://www.crunchbase.com/organization/rayzone-group> (accessed 2026-06-14).
 79. Calcalist / CTech, "The Rayzone Group's secret cyber intelligence activities revealed," 2019, <https://www.calcalistech.com/ctech/articles/0,7340,L-3884553,00.html> (named-outlet, dated: 2019 turnover of three group companies ~NIS 107M (~US\$33.25M); geolocation product *Echo*; operations in 80+ countries) (accessed 2026-06-14).
 80. Crunchbase / Tracxn / PitchBook, Septier Communication profiles (aggregators conflict — Tracxn records no raised funding; PitchBook lists ~\$1.6M — neither load-bearing; no priced valuation in public sources), <https://www.crunchbase.com/organization/septier-communication> (accessed 2026-06-14).
 81. Septier Communication Ltd., "About Septier" (company self-positioning), <https://www.septier.com/about-septier/> (self-described "fast-growing, highly profitable private company" operating in the SIGINT segment; systems deployed in 80+ countries — vendor self-statement, not an analyst assertion) (accessed 2026-06-14).
 82. Motherboard / VICE, "This Company Has Sold £1 Million Worth of Cellphone-Spying Devices to UK Police," https://motherboard.vice.com/en_us/article/pgwd5z/this-company-has-sold-pound1-million-worth-of-cellphone-spying-devices-to-uk-police (named-outlet, dated: Cellxion formed 2004; >£1M Metropolitan Police purchase 2015; UK LEA/HMRC contracts — procurement-side public footprint, no disclosed valuation) (accessed 2026-06-14).
 83. MSAB (Micro Systemation AB, Nasdaq Stockholm: MSAB B), "MSAB Interim Report – Q2 2026," dated 2026-07-16, <https://investors.msab.com/investors/newsroom/press-releases/2026/msab-interim-report-q2-2026/> (accessed 2026-07-21). Verbatim: "Net turnover amounted to 122.5 (80.5) MSEK, an increase of 52.1 per cent compared with the same period last year"; management statement "For the first time ever, we have a positive half year one in this company." Reported the company's first-ever profitable first half (net profit SEK 12.6M versus a SEK 17M loss a year earlier). Dual-dated against FY2025 net sales SEK 462M (⁷⁴), which stays as the last full-year

- actual. On 2026-09-01 MSAB announced an order “valued at SEK 25.7 million with a larger defense customer in Europe” for its XRY Pro extraction tool, with revenue “expected to be recognized in the company’s third quarter” (MSAB regulatory release, dated 2026-09-01, <https://www.inderes.dk/releases/msab-secures-a-major-contract-worth-sek-257-million-with-european-defense-organization>, accessed 2026-09-11).
84. Cellebrite, full-year 2026 guidance issued alongside the Q1 FY2026 results on 2026-05-14 – before this update window – ARR USD 567–573M (18–19% growth) and revenue USD 565–571M (19–20% growth). Cellebrite, “Cellebrite Announces First-Quarter 2026 Results,” 2026-05-14, <https://www.prnewswire.com/news-releases/cellebrite-announces-first-quarter-2026-results-302772161.html>; Form 6-K <https://www.sec.gov/Archives/edgar/data/0001854587/000121390026056220/ea029072401ex99-1.htm> (accessed 2026-07-21). Lowered on 2026-08-13 alongside the Q2 FY2026 results to ARR USD 550–560M (14–16% growth) and revenue USD 555–561M (17–18% growth), with the adjusted-EBITDA target raised to USD 153–159M – Exhibit 99.1 to Form 6-K furnished 2026-08-13, <https://www.sec.gov/Archives/edgar/data/1854587/000121390026088681/ea030188001ex99-1.htm> (accessed 2026-09-11); the 2026-05-14 guidance above is retained as the prior dated outlook. Carried at the disclosed guidance level; the ARR \$507.8M Q2 print is the current disclosed actual and the \$493M Q1 print the prior dated quarter. NASDAQ: CLBT – trading-position disclaimer applies; re-verify against the next 6-K at publication.
85. Oxygen Forensics, “3 Solutions for Mobile Forensics Challenges in 2025” (“the widespread use of end-to-end encryption by apps like WhatsApp, Signal, and Telegram makes data extraction increasingly challenging”), <https://www.oxygenforensics.com/blog/mobile-forensics-challenges-2025-solutions/> (accessed 2026-06-14); EFF, “Cell-Site Simulators / IMSI Catchers,” on the 2G-downgrade dependency for content interception, <https://sls.eff.org/technologies/cell-site-simulators-imsi-catchers> (accessed 2026-06-14).
86. European Parliamentary Research Service, “Access to data for law enforcement and digital forensics,” EPRS PE 775879 (2025), framing device-level forensic access as the law-enforcement response when communications interception is defeated by encryption, [https://www.europarl.europa.eu/RegData/etudes/BRIE/2025/775879/EPRS_BRI\(2025\)775879_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2025/775879/EPRS_BRI(2025)775879_EN.pdf) (accessed 2026-06-14).
87. Mordor Intelligence, “IMSI Catcher Market Size, Share & 2031 Growth Trends Report” (USD 207.68M in 2026; 8.08% CAGR to USD 306.47M by 2031; “5G SA-compatible interception, where no vendor has yet established a clearly dominant globally deployed commercial solution”), <https://www.mordorintelligence.com/industry-reports/imsi-catcher-market>; Mordor Intelligence, “Cellular Interception Market” (USD 695.49M (2025) → USD 735.69M (2026); 4.61% CAGR 2026–2031), <https://www.mordorintelligence.com/industry-reports/cellular-interception-market> (both accessed 2026-06-14; single-firm estimates for an opaque B2G niche – directional, not authoritative, per the thin-material discipline).
88. MarketsandMarkets, “Mobile Forensics Market” (USD 5.72B (2026) → USD 9.99B (2031); 11.8% CAGR), <https://www.marketsandmarkets.com/Market-Reports/mobile-forensics-market-29093950.html>; Mordor Intelligence, “Digital Forensics Market” (USD 14.85B (2025) → USD 31.74B (2030); 16.5% CAGR – prior edition; page re-issued 2026-07-23 and re-checked 2026-09-11: USD 7.51B (2025) → USD 8.46B (2026) → USD 15.37B (2031), 12.68% CAGR (2026–2031); “the top five vendors, OpenText, Cellebrite, Magnet Forensics, Exterro, and MSAB, accounted for about 38% of 2025 revenue” – quoted verbatim and unchanged on the re-issued page), <https://www.mordorintelligence.com/industry-reports/digital-forensics-market> (both accessed 2026-06-14; directional).
89. Cellebrite DI Ltd. (NASDAQ: CLBT), Q2 FY2026 results (quarter ended 2026-06-30, reported 2026-08-13): total revenue USD 131.1M (+16% YoY); subscription revenue USD 119.5M (+16%); ARR USD 507.8M (+21% YoY); adjusted EBITDA USD 31.8M (24.2% margin); GAAP net income USD 6.4M; 80.8% GAAP gross margin; net revenue retention 117%; full-year 2026 ARR outlook lowered the same day to USD 550–560M (14–16% growth) with the adjusted-EBITDA target raised to USD 153–159M; CEO succession announced the same day – Shiven Ramji succeeding Thomas E. Hogan, described in the company’s own Form 6-K as a planned transition, with Hogan also leaving the board. Exhibit 99.1 to Form 6-K furnished 2026-08-13, <https://www.sec.gov/Archives/edgar/data/1854587/000121390026088681/ea030188001ex99-1.htm>; CEO Form 6-K, https://www.sec.gov/Archives/edgar/data/1854587/000121390026089232/ea0302091-6k_cellebrite.htm

- (both accessed 2026-09-11). Prior dated quarter – Q1 FY2026 (quarter ended 2026-03-31, reported 2026-05-14): total revenue USD 128.3M (+19% YoY); subscription revenue USD 117.9M (+23%); ARR USD 493M (+21% YoY); adjusted EBITDA USD 30.6M (+29%); GAAP net income USD 10.9M; 82.5% GAAP gross margin; net revenue retention 115%; Q2 FY2026 ARR guidance USD 510M-513M; drone-forensics portfolio expansion. Form 6-K (earnings press-release exhibit), SEC, <https://www.sec.gov/Archives/edgar/data/0001854587/000121390026056220/ea029072401ex99-1.htm>; Cellebrite, "Cellebrite Announces First-Quarter 2026 Results," 2026-05-14, <https://www.prnewswire.com/news-releases/cellebrite-announces-first-quarter-2026-results-302772161.html> (accessed 2026-06-14). CLBT is a public issuer – the trading-position disclaimer applies. Re-verify against the next 6-K at publication; do not carry stale.
90. MSAB (Micro Systemation AB, Nasdaq Stockholm: MSAB B), "Year-end report 2025" (year ended 2025-12-31): full-year net sales SEK 462M (+14.1% YoY; +19.7% currency-adjusted); operating profit/EBIT SEK 68M (14.7% margin); profit after tax SEK 49.2M; "the most successful year in the company's history," <https://www.msab.com/updates/year-end-report-2025/> (accessed 2026-06-14).
 91. MSAB, "Mobile Forensics in 2026: The Comprehensive Guide to Extraction, Analysis, and Trends," dated 28 April 2026 ("The trend in 2026 is moving extraction to the field"; "XRY Frontline (Kiosk, Tablet, Express)" deployed "at police stations and borders"; "MSAB's latest innovation allows agencies to utilize high-power GPUs to accelerate the brute-force process. What used to take weeks can now be accomplished in hours"), <https://www.msab.com/blog/mobile-forensics-comprehensive-guide/> (accessed 2026-06-14).
 92. MSAB XRY Drone unveiled November 2025 (third-party aggregator/comparison commentary, carried as aggregator-attributed): Gitnux, "Best Cell Phone Forensics Software 2026," <https://gitnux.org/best/cell-phone-forensics-software/>; Sherlock Forensics, <https://www.sherlockforensics.com/blog/cellebrite-vs-magnet-axiom-2026.html> (accessed 2026-06-14).
 93. Cellebrite, "Cellebrite UFED I Mobile Forensic Data Extraction Tool" ("The Industry Standard for Lawfully Accessing and Collecting Digital Data"; "Extract data from anywhere, whether in the lab, a remote location, or in the field"; "UFED Touch3 Ruggedized Tablet enables comprehensive data extraction collection capabilities anywhere, whether in the lab, a remote location, or in the field"; "full file system (FFS) and physical extractions"), <https://cellebrite.com/en/products/ufed/> (accessed 2026-06-14); Cellebrite "Spring 2026 Release" (iPhone 17 / iOS 26 device access; AI-powered decoding), <https://cellebrite.com/en/products/launches-releases/spring-release-2026/> (accessed 2026-06-14); Cellebrite, "Cellebrite Brings Battlefield Intelligence to the Armed Forces with Launch of Portable Tactical Extraction Kit (C-TEK)," dated 2026-08-26 – verbatim: "a ruggedized, all-in-one mobile forensics field kit packaged in a standard-size backpack, giving military teams the hardware, software and power to access and analyze devices at the point of capture"; "Designed for units operating in disconnected environments" – <https://cellebrite.com/en/resources/press-releases/cellebrite-brings-battlefield-intelligence-to-the-armed-forces-with-launch-of-portable-tactical-extraction-kit-c-tek/> (accessed 2026-09-11).
 94. FedScoop, "DHS units to re-up contract with controversial mobile device data extractor" (U.S. CBP reported searching ~47,000 electronic devices in 2024, up from ~8,500 in 2015; DHS/ICE+HSI planned five-year IDIQ with a USD 100M ceiling for Cellebrite; CBP had eight Cellebrite contracts ending July 2025–April 2026), <https://fedscoop.com/dhs-cellebrite-privacy-drones-data-mobile-devices-ice/>; AFSC Investigate, Cellebrite contract record (ICE awarded Cellebrite at least 213 contracts worth over USD 48.6M between 2008 and April 2025), <https://investigate.afsc.org/company/cellebrite> (accessed 2026-06-14).
 95. OrangeSlices AI, "Contract Award: \$6M DOJ/FBI Cellebrite Unlock and Extraction Software" (FBI/DOJ awarded Cellebrite a contract valued at USD 5,717,319 for unlock and extraction software, start 2025-09-30, completion 2030-09-29, via the NASA SEWP contract vehicle), <https://orangeslices.ai/contract-award-6m-doj-fbi-cellebrite-unlock-and-extraction-software/> (accessed 2026-06-14).
 96. U.S. Department of Commerce, Bureau of Industry and Security – Entity List additions: NSO Group and Candiru (4 November 2021, "presumption of denial"), Federal Register, "Addition of Certain Entities to the Entity List," <https://www.federalregister.gov/documents/2021/11/04/2021-24123/addition-of-certain-entities-to-the-entity-list>; BIS press release, <https://www.bis.gov/press-release/commerce-adds-nso-group-other-foreign->

- companies-entity-list-malicious-cyber-activities; Intellexa/Cytrox added 18 July 2023, CyberScoop, "Commerce Department blacklists spyware companies Cytrox and Intellexa," <https://cyberscoop.com/commerce-department-blacklists-spyware-companies/> (accessed 2026-06-14). Descriptive market-force commentary on the never-graded remote-implant cohort; no grade, no pattern claim about any specific such vendor.
97. EU Dual-Use Regulation 2021/821, Article 5 cyber-surveillance catch-all (controlled cyber-surveillance items include "equipment for covertly extracting communication content and metadata that is transmitted over the air via wireless communication, and radiofrequency monitoring equipment"; human-rights end-use test; European Commission guidelines issued 15 October 2024). Akin Gump, "Guidelines on the Export of Cyber-surveillance Items Under Article 5 of Regulation 2021/821," <https://www.akingump.com/en/insights/alerts/guidelines-on-the-export-of-cyber-surveillance-items-under-article-5-of-regulation-2021-821>; SIPRI, "Making the most of the EU catch-all control on cyber-surveillance exports," <https://www.sipri.org/commentary/topical-background/2024/making-most-eu-catch-all-control-cyber-surveillance-exports> (accessed 2026-06-14). On the Wassenaar Category 4 (intrusion software) / Category 5 Part 1 (IP-network/RF surveillance) split: NIST/CSRC Wassenaar control-implementation summary, <https://csrc.nist.gov/> (accessed 2026-06-14).
 98. Access Now / named-outlet reporting on US sanctions actions against the Intellexa consortium (Entity List 2023-07-18; OFAC/Treasury designations Mar & Sep 2024) and the December 2025 partial lifting of sanctions on three Intellexa-linked executives — descriptive market-force color on the never-graded remote-implant surveillance cohort, named-outlet-attributed, <https://www.accessnow.org/press-release/us-new-sanctions-on-spyware-company-and-executives/> (accessed 2026-06-14).
 99. Rayzone Group, "Proximity-Data Intelligence" (re-fetched 2026-06-14): "full-scale, last-mile SIGINT capabilities"; "Tactical RF interception for 2G, 3G, 4G, and 5G networks"; "Designed for stationary, portable, and airborne deployment"; "Tactical awareness. Real-time control."; "Real-Time Situational Awareness at the Tactical Edge," <https://rayzone.com/solution/proximity-data-intelligence/>; Piranha SDR IMSI/IMEI collector and ArrowCell counter-IMSI-catcher per vendor pages and named-outlet coverage (CTech/Calcalist), <https://www.calcalistech.com/ctech/articles/0,7340,L-3884553,00.html> (accessed 2026-06-14). Private; no public 2026 financial disclosure.
 100. Septier Communication Ltd., Septier Guardian IMSI Catcher + Cellular Locator product pages (re-confirmed 2026-06-14): "capturing up to 11 different identifiers for each cellular device, including those using Bluetooth and Wi-Fi"; extracts "cellular identities, location, and other information from GSM, UMTS, LTE, and 5G devices in its coverage area"; Cellular Locator "a tactical location tracking solution that provides accurate location information for cellular devices, using advanced direction finders such as Septier Hunter," <https://www.septier.com/portfolio-item/septier-identities-catcher/> and <https://www.septier.com/portfolio-item/septier-guardian-airborne/> (accessed 2026-06-14). Private; no public 2026 financial disclosure.
 101. Cellxion (UK) cellular-interception / IMSI-catcher equipment (GX/UGX range), via the Surveillance Industry Index / named directory descriptions — named-outlet/index characterization, explicitly not vendor self-positioning, held to Wildcard caution (descriptive only, no pattern claim, no negative). Procurement-side public footprint per named-outlet reporting (Motherboard/VICE, >£1M Metropolitan Police purchase, 2015), https://motherboard.vice.com/en_us/article/pgwd5z/this-company-has-sold-pound1-million-worth-of-cellphone-spying-devices-to-uk-police (accessed 2026-06-14).
 102. Thoma Bravo and Magnet Forensics, "Magnet Forensics Announces Closing of Plan of Arrangement with Thoma Bravo," closing announcement dated 2023-04-06, <https://www.thomabravo.com/press-releases/magnet-forensics-announces-closing-of-plan-of-arrangement-with-thoma-bravo> (accessed 2026-08-22). The release confirms the Magnet arrangement closed and describes bringing Magnet Forensics and Grayshift together; it says Grayshift partnered with Thoma Bravo in 2022 but does not state a Grayshift ownership percentage. McMillan LLP, transaction counsel, "McMillan Advises Thoma Bravo in its \$1.8 Billion Acquisition of Magnet Forensics Inc.," <https://mcmillan.ca/deals-cases/mcmillan-advises-thoma-bravo-in-its-1-8-billion-acquisition-of-magnet-forensics-inc/>, supports the approximately C\$1.8B Magnet transaction value. TechCrunch, 2023-01-20, <https://techcrunch.com/2023/01/20/thoma-bravo-agrees-to-acquire-digital-forensics-firm-magnet-forensics-for-over-1b>, and GovTech, <https://www.govtech.com/biz/magnet-forensics-to-merge-with-grayshift-after-1-35b-buyout>, provide named-outlet USD-value corroboration (all accessed 2026-08-22). The C\$1.8B / USD-value

reporting belongs to Magnet's take-private, not to a purchase of Grayshift. Thoma Bravo, "Grayshift Announces Strategic Investment from Thoma Bravo," announcement dated 2022-06-21, <https://www.thomabravo.com/press-releases/grayshift-announces-strategic-investment-from-thoma-bravo> (accessed 2026-08-22). The release characterizes the transaction as a strategic growth investment, says financial terms were not disclosed, says existing investors would remain shareholders, and says the investment was subject to regulatory approvals. It does not establish majority control or a July 2022 close.

Disclosures

DISCLOSURE.

This is ProductBeacon's independent, impartial market research on the government and national-security intelligence-software market, based solely on publicly available material. No vendor sponsors, no paywalled data, no analyst-firm reuse. The author, Yohay Etsion, served as VP Product at Cognyte, which competes in fronts covered here; this report reflects the author's personal analysis, not Cognyte's position, and is not investment, legal, tax, or accounting advice. Full independence and sourcing methodology: [Methodology](#) →.

NOT ADVICE.

This report does not constitute investment, legal, tax, or accounting advice. No claim should be relied upon as the basis for any investment decision. As of September 2026, the author holds no position in any named public security, directly or via a cybersecurity-sector ETF, and is not compensated by any named vendor. Readers who use this report in investment contexts bear sole responsibility for their decisions.